

# Volatility Forecasting and Value-at-Risk Estimation using ARIMA–GARCH Models and HAR-RV Benchmark

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**Abstract:** This paper compares the GARCH family of parametric models with the semi-parametric Heterogeneous Autoregressive Realized Volatility (HAR-RV) model in the context of volatility forecasting and Value-at-Risk (VaR) estimation for daily S&P 500 returns over the period January 2010 to December 2024 (3,772 observations). Six GARCH-family specifications, formed by crossing three variance equations (GARCH, GJR-GARCH, EGARCH) with two error distributions (Gaussian and Student- $t$ ), are estimated alongside HAR-RV under an expanding-window out-of-sample scheme ( $T = 755$ ). Forecast accuracy is assessed via MSE, MAE, and QLIKE loss functions, while parametric VaR and Expected Shortfall at the 95% and 99% confidence levels are validated using the Kupiec and Christoffersen tests.

In-sample, EGARCH(1,1)-Student- $t$  achieves the lowest AIC and BIC, with estimated leverage ( $\hat{\gamma} = -0.211$ ,  $p < 0.001$ ) and degrees-of-freedom ( $\hat{\nu} = 5.49$ ) confirming asymmetric volatility response and fat-tailed innovations. Out-of-sample, HAR-RV records the lowest MSE and MAE but produces the worst QLIKE. Under the QLIKE loss function, which penalises volatility under-prediction asymmetrically, the ranking reverses completely: all six GARCH specifications outperform HAR-RV at the 0.1% significance level (DM statistics 5.37–5.83), confirming that HAR-RV’s point-forecast superiority does not extend to tail-risk calibration. In backtesting, GARCH(1,1)-Normal and EGARCH(1,1)-Normal are the only models to pass all tests at both confidence levels; however, subsample analysis reveals that their full-sample pass reflects error cancellation across regimes rather than uniform robustness, as both fail during the 2022 bear market with violation rates exceeding 9%. Student- $t$  specifications exhibit the complementary pattern, passing under stress but over-correcting in calm periods, while HAR-RV fails uniformly ( $p < 0.001$ ) due to systematic VaR under-estimation.

These findings indicate that model selection should be conditioned on the intended application: HAR-RV provides competitive point forecasts but requires distributional correction for regulatory reporting, whereas asymmetric GARCH models with Student- $t$  innovations offer the most reliable framework for tail-risk quantification under Basel III.

**Keywords:** Volatility forecasting, GARCH, GJR-GARCH, EGARCH, HAR-RV, Value-at-Risk, Expected Shortfall, Kupiec test, Christoffersen test, S&P 500.

## I. Introduction

THE accurate measurement and forecasting of conditional volatility constitutes one of the most consequential problems in empirical finance, with direct implications for derivative pricing, dynamic portfolio optimization, and the prudential management of market risk. Return volatility exhibits a number of well-documented empirical regularities that render static variance assumptions fundamentally unsuitable for risk quantification. Chief among these is *volatility clustering*, first documented by Mandelbrot [1], whereby large-magnitude returns tend to be followed by further large-magnitude returns, whilst tranquil periods likewise exhibit persistence over time.

Two paradigms have emerged to model this phenomenon. The first is the parametric GARCH family, initiated by Engle [2] and generalized by Bollerslev [3], which models conditional variance as a function of lagged squared innovations and past variances. Extensions including the GJR-GARCH [4] and EGARCH [5] accommodate the asymmetric *leverage effect*, whereby negative shocks amplify volatility more than positive shocks of equal magnitude. The second is the realized volatility paradigm, in which Andersen and Bollerslev [6] established nonparametric variance estimation from high-frequency data, and Corsi [7] proposed the Heterogeneous Autoregressive model (HAR-RV) to exploit the multi-horizon persistence structure implied by the Heterogeneous Market Hypothesis [8].

The operational significance of volatility forecasting is most immediate in market risk measurement. The Value-at-Risk (VaR) metric quantifies the maximum loss not exceeded with a prescribed probability, but Artzner et al. [9] showed that VaR violates sub-additivity and proposed Expected Shortfall (ES) as a coherent alternative. ES has since been adopted by the Basel Committee as the primary regulatory risk measure under the Fundamental Review of the Trading Book [10]. The formal validation of VaR and ES through the Proportion of Failures test of Kupiec [11] and the Conditional Coverage framework of Christoffersen [12] constitutes a binding regulatory requirement under Basel III.

Notwithstanding the extensive literature on each component, a systematic comparison of the complete ARIMA-GARCH pipeline against the HAR-RV benchmark within a unified expanding-window framework, extended to formal VaR/ES backtesting for both paradigms, has received limited attention. This study addresses that gap by evaluating seven competing specifications on daily S&P 500 returns spanning January 2010 to December 2024, a fifteen-year sample that encompasses two stress episodes: the COVID-19-induced dislocation of 2020 and the monetary-policy-driven bear market of 2022.

### A. Research Objectives

The present study is organized around the following four research questions:

- 1) Among the GARCH-family specifications considered (GARCH, GJR-GARCH, and EGARCH), which model, paired with which conditional error distribution (Gaussian or Student- $t$ ), yields the most parsimonious in-sample fit to the S&P 500 daily return series, as assessed by the Akaike and Bayesian information criteria?
- 2) Does the HAR-RV model generate statistically superior out-of-sample conditional volatility forecasts relative to the best-fitting GARCH specification, as evaluated under the MSE, MAE, and QLIKE loss functions over the hold-out sample?
- 3) Do the parametric VaR estimates produced by each model satisfy the statistical requirements of correct unconditional coverage and violation independence, as assessed by the Kupiec [11] and Christoffersen [12] tests at both the 95% and 99% confidence levels?
- 4) To what extent does substituting the Gaussian with the Student- $t$  distribution improve VaR calibration accuracy, particularly in the extreme quantiles of the loss distribution?

### B. Contributions

The present paper advances the existing literature along three dimensions. First, it conducts a rigorous side-by-side evaluation of parametric GARCH and semi-parametric HAR-RV paradigms within a unified expanding-window framework that precludes look-ahead bias. Second, it extends the standard VaR backtesting apparatus to encompass HAR-RV-implied risk estimates, a dimension that receives limited treatment in the existing literature. Third, by exploiting a fifteen-year sample spanning two distinct stress regimes, it furnishes evidence on the relative robustness of each model class under conditions of acute market stress, which is of direct relevance to practitioners engaged in risk model validation and regulatory capital adequacy assessment.

### C. Paper Structure

The remainder of this paper is organized as follows. Section II surveys the relevant literature on GARCH-family models, realized volatility, and backtesting frameworks. Section III describes the data and presents summary statistics. Section IV develops the methodological framework. Section V presents the empirical results. Section VI concludes.

## II. Literature Review

This section surveys the theoretical and empirical foundations underpinning the models and methods employed in the present study. The review covers four thematic areas: the ARCH/GARCH family of conditional variance models and their error distributions, the HAR-RV framework for realized volatility forecasting, the Value-at-Risk and Expected Shortfall paradigm together with its regulatory context, and the research gap that motivates the current analysis.

### A. ARCH/GARCH Models and Their Extensions

The formal econometric treatment of time-varying conditional variance originates with Engle [2], who introduced the Autoregressive Conditional Heteroskedasticity (ARCH) model. Engle demonstrated that the conditional variance of a financial return series can be expressed as a function of lagged squared residuals, providing a rigorous framework for

the volatility clustering phenomenon first documented by Mandelbrot [1], whereby large-magnitude returns tend to be followed by further large-magnitude returns regardless of sign.

The principal limitation of the ARCH model, its requirement for a long lag structure to adequately represent volatility persistence, was resolved by Bollerslev [3], who proposed the Generalized ARCH (GARCH) model. By augmenting the variance equation with lagged conditional variances in addition to lagged squared innovations, the GARCH(1,1) specification achieves a parsimonious representation of volatility dynamics that has since become the *de facto* benchmark in the empirical volatility literature. The degree of shock persistence, measured by the sum  $\alpha + \beta$ , is consistently found to lie close to but below unity for major equity indices, implying mean-reverting yet long-lived volatility [3].

A further empirical regularity that symmetric GARCH does not capture is the leverage effect: negative return shocks generate a disproportionately larger increase in subsequent volatility than positive shocks of equal magnitude. To accommodate this asymmetry, Glosten et al. [13] proposed the GJR-GARCH model, which augments the variance equation with an indicator function that activates only for negative innovations. The associated coefficient  $\gamma$  directly quantifies the leverage effect, and a positive, statistically significant  $\gamma$  constitutes evidence that bad news amplifies variance beyond the symmetric GARCH response. Independently, Nelson [5] introduced the Exponential GARCH (EGARCH) model, which parameterises the logarithm of conditional variance, guaranteeing positivity without explicit parameter constraints and accommodating asymmetric volatility responses through a sign term. Both GJR-GARCH and EGARCH have been shown to produce superior in-sample fits relative to the symmetric GARCH benchmark for major equity indices [5, 13].

## B. Error Distributions in GARCH Models

The Gaussian assumption for standardised residuals, while analytically convenient, is widely recognised as inadequate for financial returns. Mandelbrot [1] was among the first to document that the empirical return distribution exhibits heavier tails than the normal distribution implies, meaning that extreme losses occur far more frequently than a Gaussian model would predict. Bollerslev [14] addressed this limitation by proposing the Student- $t$  distribution as the conditional error distribution within the GARCH framework. The degrees-of-freedom parameter  $\nu$  governs tail thickness and is estimated jointly with the variance parameters. Values of  $\nu < 10$  are generally interpreted as evidence of economically meaningful excess tail risk, which translates directly into more conservative and better-calibrated VaR estimates at high confidence levels. Empirical studies consistently confirm that GARCH- $t$  models outperform their Gaussian counterparts in VaR accuracy, especially at the extreme quantiles most relevant for risk management [14].

## C. Realized Volatility and the HAR-RV Model

A methodologically distinct approach to volatility measurement and forecasting has developed around the concept of realized volatility (RV), a nonparametric estimator constructed from high-frequency intraday data. Andersen and Bollerslev [15] established the theoretical foundations of RV as a consistent estimator of the underlying quadratic variation, showing that the sum of squared intraday returns converges to the true integrated variance as the sampling frequency increases.

The principal forecasting model within this tradition is the Heterogeneous Autoregressive model of Realized Volatility (HAR-RV), introduced by Corsi [7]. The HAR-RV is grounded in the Heterogeneous Market Hypothesis of Müller et al. [8], which posits that financial markets are populated by participants operating at heterogeneous investment horizons. The model operationalises this cascade structure by regressing next-period realized volatility on realized volatility aggregated over daily, weekly, and monthly horizons, estimated by ordinary least squares. Despite its structural simplicity, this multi-component design achieves competitive out-of-sample forecasting performance relative to more elaborate parametric models [7].

When high-frequency intraday data are unavailable, a standard practice is to proxy realized volatility using the absolute daily return  $|r_t|$ , following the approach of Ding et al. [16], who documented the long-memory autocorrelation structure of absolute returns as an estimator of latent volatility. However,  $|r_t|$  is a noisy estimator of true integrated variance that discards all information about intraday price movements.

## D. Value-at-Risk, Expected Shortfall, and Backtesting

Value-at-Risk (VaR) quantifies the maximum portfolio loss not exceeded with a given probability over a specified holding period, and has been the dominant market risk measure in regulatory practice since its adoption under the original Basel Accord. Artzner et al. [9] identified a fundamental theoretical shortcoming of VaR: it fails to satisfy the

sub-additivity axiom required of coherent risk measures. As a coherent alternative, Artzner et al. proposed Expected Shortfall (ES), which measures the expected loss conditional on the loss exceeding the VaR threshold. The Basel Committee on Banking Supervision formalised the transition from VaR to ES when it adopted ES at the 97.5% confidence level as the primary market risk metric under the Fundamental Review of the Trading Book [10].

The formal validation of VaR through backtesting constitutes both an academic requirement and a binding regulatory obligation. The Proportion of Failures (POF) test of Kupiec [11] provides a likelihood ratio statistic for the null hypothesis that the empirical violation rate equals the nominal confidence level. This limitation was extended by Christoffersen [12], whose Conditional Coverage (CC) framework jointly tests for correct unconditional coverage and independence of violations by modelling the exceedance indicator sequence as a first-order Markov chain. Models that pass the Kupiec test but fail the Christoffersen test are particularly problematic in practice, as they give a false impression of adequate calibration while systematically underestimating risk precisely when it is most consequential.

## E. Research Gap and Motivation

The foregoing review establishes that the GARCH family and the HAR-RV constitute the two dominant paradigms for conditional volatility forecasting. The broad consensus in the comparative literature, as exemplified by Andersen and Bollerslev [15] and Corsi [7], is that HAR-RV outperforms GARCH-family models when high-frequency intraday data are available. However, this consensus rests on three assumptions that the present study explicitly relaxes, and the failure to relax them materially limits the practical applicability of existing findings.

**First**, existing comparative studies treat realized volatility as an observable, precisely measured quantity constructed from tick-by-tick transaction data. This assumption does not hold in the large majority of real-world risk management settings, where practitioners work exclusively with daily closing prices. While Ding et al. [16] established that absolute daily returns  $|r_t|$  carry genuine long-memory information about latent volatility, no systematic study has evaluated how the HAR-RV performs relative to the GARCH family specifically under this noisier proxy. The distinction matters because the two paradigms respond very differently to measurement noise: GARCH models filter noise through their recursive variance equation and update estimates each period, whereas HAR-RV aggregates noise directly into the regression without any such filtering mechanism. Whether this difference favours one paradigm over the other, and whether it changes across loss functions that weight forecast errors asymmetrically, is an open empirical question that the existing literature does not cleanly answer.

**Second**, the comparative literature has focused almost exclusively on point forecast accuracy under symmetric loss functions such as MSE and MAE. This focus obscures the dimension that matters most in applied risk management: whether the model’s implied loss distribution is correctly calibrated in its tails. Papers that find HAR-RV superior under MSE [7] do not address whether its VaR estimates would satisfy the Kupiec and Christoffersen requirements that are legally binding under Basel III [10]. Specifically, forecast accuracy and regulatory VaR validity need not agree, and the failure to evaluate both within a unified framework means that existing comparative conclusions provide only partial guidance for practitioners who must simultaneously optimise both objectives. The present study closes this gap by applying an identical backtesting protocol to both paradigms and examining whether any model that excels at point forecasting also satisfies the regulatory coverage requirements.

**Third**, the robustness of these competing paradigms under market stress has received insufficient attention. Most existing comparative studies are conducted over relatively homogeneous, low-volatility sample periods that do not distinguish between fundamentally different types of market dislocation. The COVID-19 market crash of March 2020, which produced a single-day S&P 500 return of  $-12.77\%$  representing approximately 11.7 unconditional standard deviations, and the sustained bear market of 2022, driven by the most aggressive Federal Reserve rate-hiking cycle in four decades, represent qualitatively different stress regimes. The former is an acute, short-lived shock that tests a model’s ability to react rapidly to an extreme outlier; the latter is a slow-moving, persistent drawdown that tests its ability to sustain elevated variance forecasts over many months without mean-reverting prematurely. A model that performs adequately across both failure modes provides substantially stronger evidence of robustness than one evaluated on a homogeneous sample. No existing study exploits a sample that cleanly separates these two failure modes within a single unified framework.

The present study addresses all three dimensions simultaneously: it evaluates both paradigms with the same data-constrained proxy used in practice, applies an identical forecast accuracy and regulatory backtesting protocol to both, and does so over a fifteen-year sample that encompasses both acute and persistent stress regimes. This design allows conclusions that are relevant to data-constrained practitioners, regulatory compliance officers, and researchers concerned with model robustness under stress, three audiences whose requirements existing studies address only partially.

and in isolation.

### III. Data

#### A. Data Source and Description

The empirical analysis is conducted on daily closing prices of the S&P 500 Index (ticker: ^GSPC), retrieved from Yahoo Finance via the `yfinance` Python library. The sample spans from January 5, 2010 to December 30, 2024, yielding a total of 3,772 trading-day observations after removing the first observation lost to differencing. The S&P 500 is chosen as the empirical subject for two reasons. First, it is the most widely followed benchmark for U.S. equity market performance, making it the canonical test case for volatility models in the academic literature. Second, the fifteen-year sample window deliberately encompasses two episodes of acute market stress – the COVID-19-induced dislocation of early 2020 and the monetary-policy-driven bear market of 2022 – which provide a demanding out-of-sample evaluation environment for the candidate models.

The primary variable used throughout the analysis is the continuously compounded daily log return, defined as:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right) \times 100, \quad (1)$$

where  $P_t$  denotes the adjusted closing price on day  $t$  and the multiplication by 100 expresses returns in percentage terms. Log returns are preferred over simple returns for three reasons: they are time-additive over multi-period horizons, they are more likely to be symmetrically distributed, and they remove the upward trend in raw prices that renders the price series non-stationary and therefore unsuitable for direct modelling.

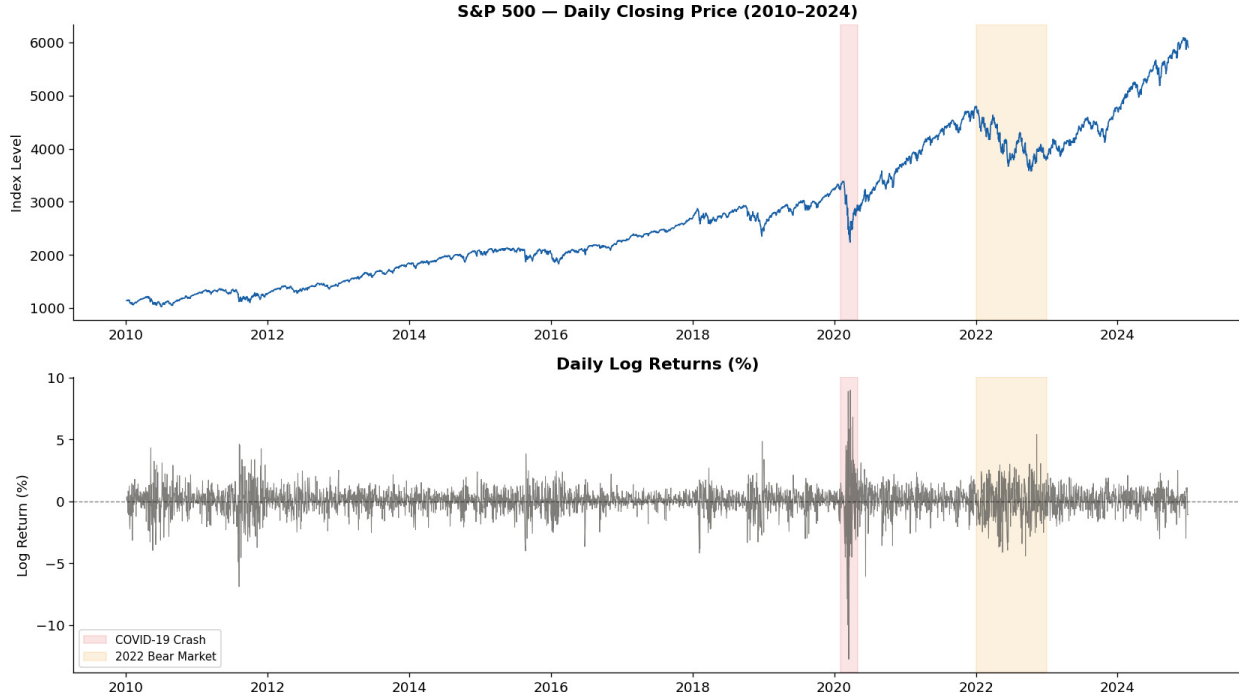
Figure 1 provides a visual overview of both series. The upper panel displays the index level over the full sample, growing from approximately 1,100 in January 2010 to nearly 6,000 by end-2024, with the two stress episodes highlighted in shading. The lower panel plots daily log returns and makes the key empirical regularities immediately visible: the return series fluctuates around zero with no discernible trend, but its dispersion is clearly not constant over time. Long stretches of relatively small returns are punctuated by brief episodes of extreme volatility, most visibly in the COVID-19 crash of March 2020 and the elevated turbulence of 2022. This pattern of *volatility clustering* is the primary empirical motivation for the GARCH family of models.

#### B. Descriptive Statistics

Table 1 reports the summary statistics for the full sample of log returns. Several features of the distribution are immediately apparent and are directly relevant to the model selection decisions made in subsequent sections.

**Table 1 Descriptive Statistics of S&P 500 Daily Log Returns (January 2010 – December 2024). Returns are expressed in percentage terms. The Jarque-Bera statistic tests the null hypothesis of normality.**

| Statistic              | Value     |
|------------------------|-----------|
| Observations           | 3,772     |
| Mean (%)               | 0.0438    |
| Standard Deviation (%) | 1.0890    |
| Minimum (%)            | −12.7652  |
| 25th Percentile (%)    | −0.3810   |
| Median (%)             | 0.0681    |
| 75th Percentile (%)    | 0.5675    |
| Maximum (%)            | 8.9683    |
| Skewness               | −0.7262   |
| Excess Kurtosis        | 13.2144   |
| Jarque-Bera Statistic  | 27,696.63 |
| Jarque-Bera $p$ -value | < 0.001   |



**Fig. 1 S&P 500 Daily Closing Price and Log Returns (January 2010 – December 2024).** The upper panel shows the index level; the lower panel shows daily log returns (in %). Pink shading indicates the COVID-19 crash period (Q1 2020); orange shading indicates the 2022 bear market. Volatility clustering is clearly visible in the lower panel: large-magnitude returns concentrate in the two shaded episodes, while the majority of the sample exhibits comparatively low dispersion.

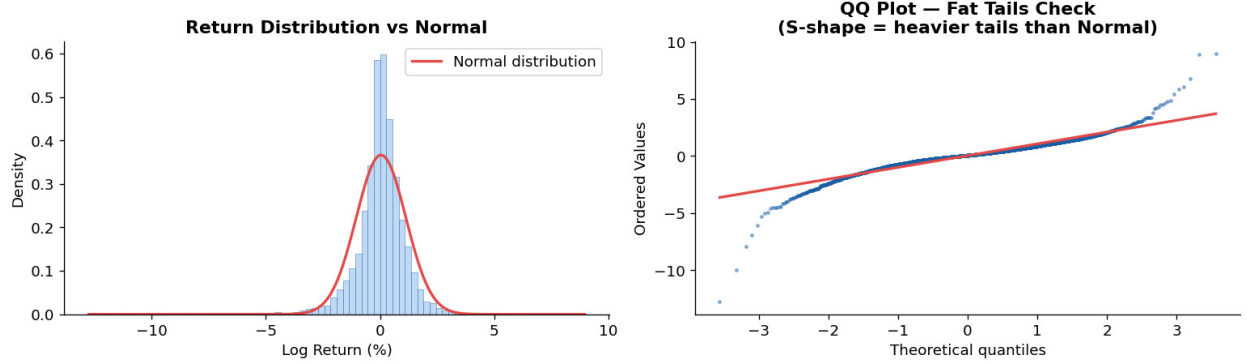
The mean daily log return of 0.0438% is close to zero, consistent with the near-martingale behavior expected of daily equity returns. The standard deviation of 1.0890% represents the unconditional, average level of daily risk, but as discussed below, this figure conceals substantial time-variation in the second moment.

The negative skewness of  $-0.7262$  indicates that the left tail of the return distribution is heavier than the right tail: large losses occur more frequently than gains of equal magnitude. This asymmetry is one of the best-documented stylized facts of equity returns and has a natural interpretation in terms of the leverage effect, whereby falling equity prices raise a firm's financial leverage and thereby amplify subsequent return volatility.

The excess kurtosis of 13.2144 is the most diagnostic statistic for model selection purposes. A value this large – more than four times the kurtosis of a normal distribution – implies that extreme return realizations, both positive and negative, occur far more frequently than a Gaussian model would assign non-negligible probability. This is precisely the *fat-tail* property documented by Mandelbrot [1]. In practical terms, it means that the largest losses in the sample, such as the  $-12.77\%$  daily return recorded during the COVID-19 crash of March 2020, would be assigned a probability of effectively zero under the Gaussian assumption. The Jarque-Bera statistic of 27,696.63 (with  $p < 0.001$ ) formally rejects the null hypothesis of normality at any conventional significance level. Together, these findings provide a strong a priori justification for employing the Student- $t$  distribution rather than the Gaussian assumption in the GARCH variance equations, and they motivate the comparative evaluation of both distributional assumptions that is central to the present study.

Figure 2 provides a visual confirmation of these departures from normality. The left panel superimposes a fitted Gaussian density on the histogram of log returns: the empirical distribution is markedly more peaked at the center (leptokurtic) and has noticeably heavier tails on both sides, with several observations extending well beyond  $\pm 5\%$  that would be essentially impossible under the fitted normal curve. The right panel presents a normal quantile-quantile (QQ) plot, in which perfect normality would produce a straight line. The pronounced S-shape deviation confirms heavy tails in both directions: the most extreme negative returns (bottom-left) and positive returns (top-right) lie far above and below the reference line, respectively. This S-shape is the graphical signature of a leptokurtic distribution and provides

further visual support for the use of fat-tailed conditional distributions in the GARCH specifications.



**Fig. 2** Return Distribution and Normal QQ Plot for S&P 500 Daily Log Returns (January 2010 – December 2024). *Left panel:* Histogram of log returns (blue bars) with a fitted Normal density overlay (red curve). The empirical distribution is more peaked and has heavier tails than the Normal, particularly on the left side. *Right panel:* Normal QQ plot. Deviations from the 45-degree reference line (red) indicate departures from normality. The characteristic S-shape confirms fat tails in both the lower and upper extremes of the distribution. Skewness =  $-0.7262$ ; Excess Kurtosis =  $13.2144$ ; Jarque-Bera  $p < 0.001$ .

### C. Train/Test Split

The dataset is partitioned into a training set and a test set following a strictly chronological split, with no shuffling of observations. This approach is essential for time series data: any shuffling or random sampling would introduce look-ahead bias by allowing the model access to future information during estimation, which would produce artificially optimistic out-of-sample performance statistics.

An 80/20 split is applied, yielding a training set of 3,017 observations covering January 5, 2010 to December 28, 2021, and a test set of 755 observations covering December 29, 2021 to December 30, 2024. The training set is used for in-sample parameter estimation and model selection via the Akaike and Bayesian information criteria. The test set serves as the out-of-sample evaluation window for volatility forecast accuracy, VaR calibration, and formal backtesting. All forecast evaluation is conducted using an expanding-window scheme, described in detail in Section V, to ensure that each one-step-ahead forecast uses only information available up to the forecast origin.

### D. Major Market Events in the Sample

The fifteen-year sample period contains a rich cross-section of market regimes that provides a demanding test of the candidate models. Table 2 summarizes the principal episodes.

The COVID-19 episode is particularly important for the evaluation of VaR models. The daily return of  $-12.77\%$  recorded on March 16, 2020 represents a realization that is approximately 11.7 standard deviations below the sample mean under the Gaussian assumption, an event with a theoretical probability of effectively zero under normality. This single observation, along with the cluster of large negative returns in March 2020, is a direct empirical demonstration of why fat-tailed conditional distributions are necessary for credible risk measurement.

The 2022 bear market, by contrast, was a slower-moving but persistent stress episode driven by the Federal Reserve’s most aggressive rate-hiking cycle in four decades. Unlike the sharp V-shaped shock of 2020, the volatility of 2022 was elevated and sustained over several months, which tests the ability of each model to capture persistent volatility regimes through its variance dynamics. Together, these two episodes represent qualitatively different types of market stress and thereby provide a comprehensive evaluation environment for assessing the robustness of the candidate volatility models and their implied VaR estimates.

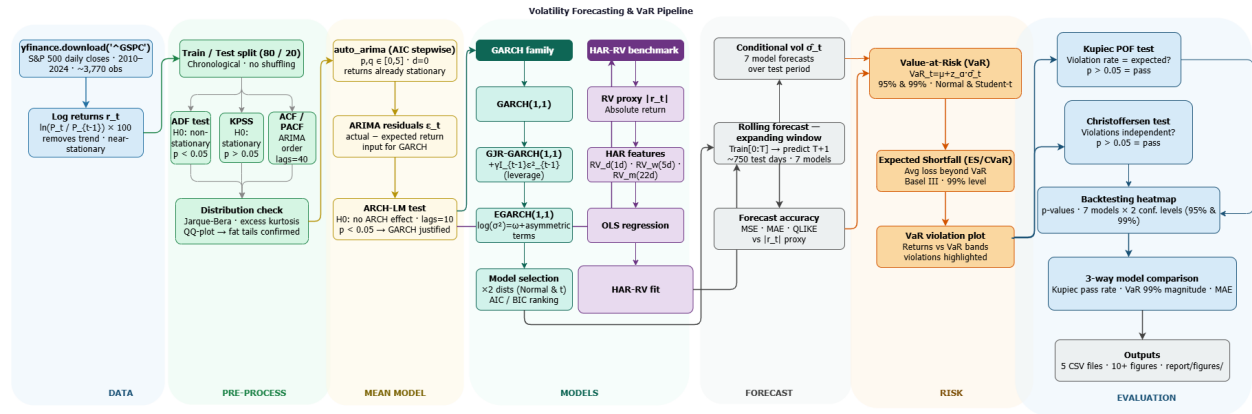
## IV. Methodology

This section presents the complete modelling pipeline employed in the present study. The pipeline proceeds sequentially through seven stages: (1) log-return construction and stationarity testing, (2) ARIMA mean equation

**Table 2 Major Market Events in the S&P 500 Sample Period (January 2010 – December 2024). Approximate peak-to-trough drawdowns are reported where applicable.**

| Period     | Event                                                                      | Volatility Regime                         |
|------------|----------------------------------------------------------------------------|-------------------------------------------|
| 2010–2019  | Sustained bull market; European debt crisis (2011); Flash Crash (May 2010) | Mostly low, with episodic spikes          |
| 2020 Q1    | COVID-19 crash (March 2020); –12.77% single-day return                     | Extreme: single-day return $> 10\sigma$   |
| 2020 Q2–Q4 | V-shaped recovery driven by fiscal and monetary stimulus                   | Elevated but declining                    |
| 2022       | Fed rate-hiking cycle; S&P 500 bear market ( $> 20\%$ drawdown)            | High and persistent; clustered violations |
| 2023–2024  | Recovery and new all-time highs                                            | Low to moderate                           |

specification, (3) the ARCH-LM diagnostic test, (4) estimation of the six GARCH-family models, (5) the HAR-RV benchmark specification, (6) expanding-window out-of-sample forecasting, and (7) VaR/ES Computation together with



**Fig. 3 Workflow pipeline for the present study.**

formal backtesting. Figure 3 summarizes this workflow.

### A. Log Returns and Stationarity Testing

As established in Section III, the primary modelling variable is the daily log return  $r_t = \ln(P_t/P_{t-1}) \times 100$ . Before fitting any time-series model it is necessary to verify that the return series is stationary, meaning that its mean, variance, and autocovariance structure do not change over time. A non-stationary series would invalidate the asymptotic properties of all subsequent estimators.

Two complementary tests are applied, chosen because they test opposite null hypotheses and therefore provide a more robust joint assessment than either test alone.

The **Augmented Dickey-Fuller (ADF)** test [17] has the null hypothesis  $H_0$ : the series contains a unit root (non-stationary). The test statistic is:

$$\Delta y_t = \alpha + \beta t + \gamma y_{t-1} + \sum_{i=1}^k \delta_i \Delta y_{t-i} + \varepsilon_t, \quad (2)$$

where  $\gamma = 0$  under the null. Rejecting  $H_0$  (i.e.  $p < 0.05$ ) provides evidence of stationarity.



The **KPSS** test [18] has the null hypothesis  $H_0$ : the series is stationary. Failure to reject  $H_0$  (i.e.  $p \geq 0.05$ ) is consistent with stationarity. Together, rejecting ADF and failing to reject KPSS constitutes the strongest possible combined evidence that the series is  $I(0)$  stationary.

## B. ARIMA Mean Equation

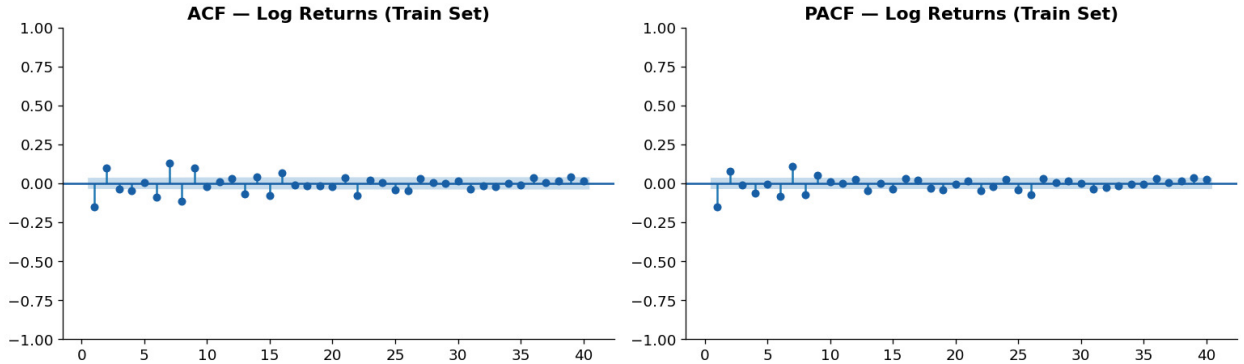
Volatility models such as GARCH are applied to the residuals of a mean equation rather than to raw returns directly. This ensures that any predictable linear structure in the conditional mean is removed first, so that the GARCH model captures only the dynamics of the conditional variance. The mean equation takes the general ARIMA( $p, d, q$ ) form:

$$r_t = \mu + \sum_{i=1}^p \phi_i r_{t-i} + \sum_{j=1}^q \theta_j \varepsilon_{t-j} + \varepsilon_t, \quad \varepsilon_t \sim (0, \sigma_t^2), \quad (3)$$

where  $\mu$  is the unconditional mean,  $\phi_i$  are the autoregressive (AR) coefficients governing the dependence of  $r_t$  on its own past values,  $\theta_j$  are the moving-average (MA) coefficients governing the dependence on past innovations, and  $\varepsilon_t$  is the mean-zero innovation sequence with time-varying conditional variance  $\sigma_t^2$ . Since log returns are already  $I(0)$  stationary (as confirmed by the stationarity tests), the integration order is fixed at  $d = 0$  throughout.

The optimal lag orders ( $p, q$ ) are selected via a stepwise search over the AIC criterion using the `auto_arima` function from the `pmdarima` library, with maximum search bounds of  $p, q \leq 5$ . The AR coefficients  $\phi_i$  and MA coefficients  $\theta_j$  capture any weak autocorrelation in the return series, and the resulting residuals  $\hat{\varepsilon}_t$  constitute the input to the GARCH variance models.

Figure 4 displays the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) of the training-set log returns up to 40 lags. The interpretation of these plots follows a standard diagnostic rule: a sharp cut-off in the PACF after lag  $p$  suggests an AR( $p$ ) process, while a sharp cut-off in the ACF after lag  $q$  suggests an MA( $q$ ) process. In both panels, virtually all bars lie within the 95% confidence band (shaded region), indicating that the autocorrelation in the level of returns is economically negligible. The absence of strong linear structure confirms that daily log returns are very close to serially uncorrelated, which is consistent with the efficient market hypothesis at the daily frequency. The `auto_arima` stepwise search, which formally minimizes AIC, selected ARIMA(4, 0, 0) as the best-fitting mean specification.



**Fig. 4** Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) of S&P 500 daily log returns on the training set (January 2010 – December 2021,  $T = 3,017$  observations). The shaded blue band represents the 95% confidence interval under the null hypothesis of zero autocorrelation. *Left panel:* ACF up to 40 lags. *Right panel:* PACF up to 40 lags. Nearly all bars lie within the confidence band, indicating that log returns exhibit negligible linear serial dependence. The selected ARIMA order is (4, 0, 0).

## C. ARCH Effect Test

Before fitting a GARCH model, it is necessary to verify that the ARIMA residuals  $\hat{\varepsilon}_t$  actually exhibit conditional heteroskedasticity, that is, that their variance is not constant over time. This is established using the ARCH Lagrange Multiplier (ARCH-LM) test proposed by Engle [2].

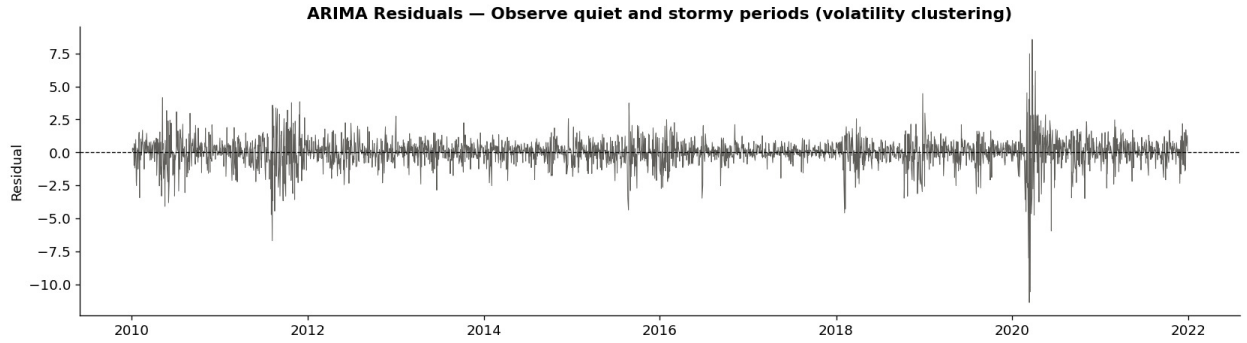
The test regresses the squared residuals on their own lags:

$$\hat{\varepsilon}_t^2 = a_0 + \sum_{i=1}^m a_i \hat{\varepsilon}_{t-i}^2 + u_t, \quad (4)$$

and tests  $H_0 : a_1 = a_2 = \dots = a_m = 0$  (no ARCH effect, i.e. constant variance). The LM statistic  $T \cdot R^2$  follows a  $\chi^2(m)$  distribution asymptotically under  $H_0$ , where  $m$  is the number of lags. In this study,  $m = 10$  lags are used. Rejecting  $H_0$  confirms that the squared residuals are serially correlated, which means that the variance is not constant and that a GARCH-type model is statistically warranted.

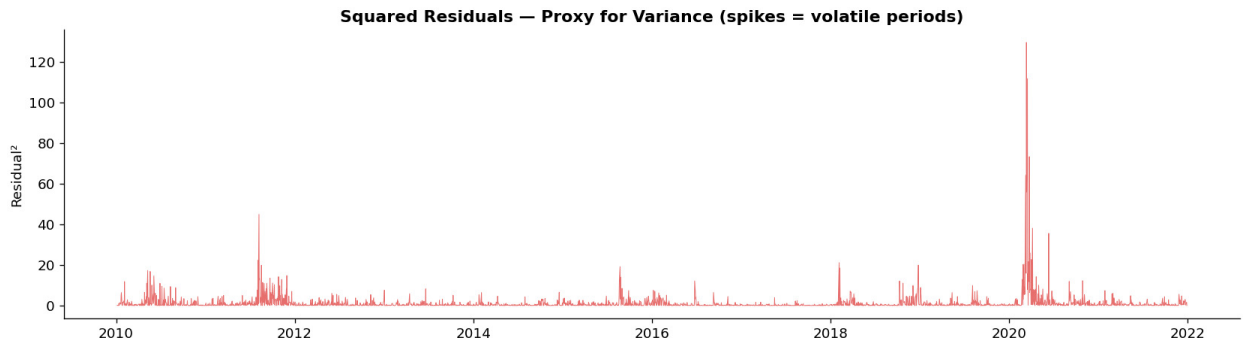
Figures 5–7 provide a visual three-panel diagnostic of the ARIMA(4, 0, 0) residuals that complements the formal ARCH-LM test.

Figure 5 plots the ARIMA residuals over the training period. The series fluctuates around zero with no discernible trend, confirming that the mean equation has been adequately specified. However, the dispersion of the residuals is clearly non-constant: stretches of small residuals in low-volatility regimes alternate with episodes of large positive and negative spikes, most prominently during the COVID-19 crash of March 2020. This visual pattern of *volatility clustering* in the residuals is the direct motivation for applying a GARCH variance model.



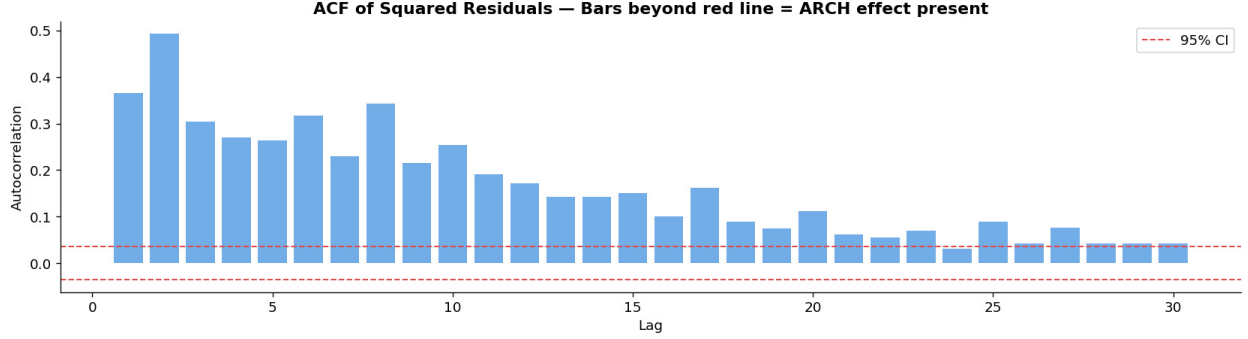
**Fig. 5** ARIMA(4, 0, 0) residuals over the training set (January 2010 – December 2021). The residuals fluctuate around zero (confirming adequate mean specification) but exhibit clear heteroskedasticity: tranquil periods of small residuals alternate with volatile episodes, particularly the COVID-19-induced spike in early 2020 (residual exceeding  $\sim 10\%$ ). This volatility clustering pattern motivates GARCH modelling of the conditional variance.

Figure 6 plots the squared residuals  $\hat{\varepsilon}_t^2$ , which serve as a proxy for daily variance. The dominance of a single spike near 2020 (squared residual exceeding 125) makes the heteroskedasticity in the variance process unmistakable: the squared residuals are far from the constant level that would be expected under homoskedasticity, and their behavior is clearly driven by time-varying volatility regimes.



**Fig. 6** Squared ARIMA(4, 0, 0) residuals over the training set (January 2010 – December 2021), serving as a proxy for daily variance. The extreme spike in early 2020 (reaching approximately 128) corresponds to the COVID-19 market dislocation and dwarfs all other variance episodes in the sample. The pronounced non-constancy of the squared residuals confirms the presence of conditional heteroskedasticity.

Figure 7 plots the ACF of the squared residuals up to 30 lags. In sharp contrast to the ACF of the level returns (Figure 4), virtually every bar in this plot extends far beyond the 95% confidence band (red dashed line), with autocorrelations at short lags exceeding 0.50. This strong and slowly decaying serial dependence in the squared residuals is the defining signature of an ARCH process and constitutes graphical confirmation of the ARCH-LM test result. The slow decay also suggests that volatility shocks are highly persistent, which motivates the use of the GARCH(1,1) specification rather than a finite-order ARCH model.



**Fig. 7** ACF of squared ARIMA(4,0,0) residuals (training set, up to 30 lags). The red dashed lines mark the 95% confidence interval under the null of zero autocorrelation. All bars at short lags substantially exceed the confidence band (autocorrelation at lag 2 exceeds 0.50), confirming strong and persistent serial dependence in the squared residuals. This pattern is consistent with an ARCH process and statistically justifies the use of GARCH-family models for the conditional variance.  $\text{ARCH-LM}(\chi^2) = 982.20, p < 0.001$ .

#### D. GARCH-Family Models

Six GARCH-family models are estimated, formed by the full factorial combination of three variance specifications and two conditional error distributions, as summarized in Table 3.

**Table 3** Summary of All Models Estimated. The seven candidate models comprise six GARCH-family specifications and the HAR-RV benchmark. For GARCH models,  $(\omega, \alpha, \beta, \gamma, \nu)$  denote the intercept, ARCH coefficient, GARCH coefficient, asymmetry parameter, and Student- $t$  degrees of freedom, respectively. For HAR-RV,  $(c, \beta_d, \beta_w, \beta_m)$  denote the intercept and the daily, weekly, and monthly coefficients.

| Model          | Distribution | Parameters                           | Key Feature              |
|----------------|--------------|--------------------------------------|--------------------------|
| GARCH(1,1)     | Normal       | $\omega, \alpha, \beta$              | Symmetric variance       |
| GARCH(1,1)     | Student- $t$ | $\omega, \alpha, \beta, \nu$         | Fat tails                |
| GJR-GARCH(1,1) | Normal       | $\omega, \alpha, \beta, \gamma$      | Leverage effect          |
| GJR-GARCH(1,1) | Student- $t$ | $\omega, \alpha, \beta, \gamma, \nu$ | Leverage + fat tails     |
| EGARCH(1,1)    | Normal       | $\omega, \alpha, \beta, \gamma$      | Log-variance, asymmetric |
| EGARCH(1,1)    | Student- $t$ | $\omega, \alpha, \beta, \gamma, \nu$ | Log-variance + fat tails |
| HAR-RV (OLS)   | —            | $c, \beta_d, \beta_w, \beta_m$       | Multi-horizon RV         |

##### 1. GARCH(1,1)

The baseline model is the symmetric GARCH(1,1) of Bollerslev [3], whose variance equation is:

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2, \quad (5)$$

where  $\omega > 0, \alpha \geq 0, \beta \geq 0$ , and the covariance-stationarity constraint  $\alpha + \beta < 1$  must hold. The ARCH coefficient  $\alpha$  measures the sensitivity of current variance to the most recent squared shock: a larger  $\alpha$  implies that a sudden spike

in returns today has a stronger and more immediate effect on tomorrow's perceived risk. The GARCH coefficient  $\beta$  measures persistence: a value close to one implies that elevated variance decays very slowly, consistent with the long-lived volatility swings observed in equity markets. The sum  $(\alpha + \beta)$  is the total persistence parameter; values close to, but below, unity imply that shocks die out gradually rather than permanently.

## 2. GJR-GARCH(1,1)

The GJR-GARCH model of Glosten, Jagannathan, and Runkle [4] augments the GARCH variance equation with an indicator function to capture the leverage effect:

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \gamma \varepsilon_{t-1}^2 \mathbf{1}[\varepsilon_{t-1} < 0] + \beta \sigma_{t-1}^2, \quad (6)$$

where  $\mathbf{1}[\varepsilon_{t-1} < 0]$  equals one when the lagged innovation is negative (bad news) and zero otherwise. The parameter  $\gamma$  directly quantifies the leverage effect: a positive and statistically significant  $\gamma$  confirms that negative return shocks amplify variance by  $(\alpha + \gamma)$ , compared with only  $\alpha$  for a positive shock of equal size. This asymmetric response reflects the empirical observation that falling equity prices raise a firm's financial leverage and heighten investor uncertainty more than gains of the same magnitude.

## 3. EGARCH(1,1)

Nelson's [5] Exponential GARCH model parameterizes the logarithm of conditional variance:

$$\ln \sigma_t^2 = \omega + \beta \ln \sigma_{t-1}^2 + \alpha \left( \frac{|\varepsilon_{t-1}|}{\sigma_{t-1}} - \sqrt{\frac{2}{\pi}} \right) + \gamma \frac{\varepsilon_{t-1}}{\sigma_{t-1}}, \quad (7)$$

where the term  $|\varepsilon_{t-1}|/\sigma_{t-1}$  captures the magnitude of the standardized shock and the sign term  $\gamma \varepsilon_{t-1}/\sigma_{t-1}$  captures the direction. A negative value of  $\gamma$  implies that negative standardized shocks raise log-variance by more than positive shocks of equal magnitude, which is the EGARCH representation of the leverage effect. The log-variance formulation has two practical advantages over the level-variance GARCH: it guarantees  $\sigma_t^2 > 0$  for any parameter values without imposing explicit non-negativity constraints, and it can represent more complex asymmetric shapes in the news impact curve.

## 4. Conditional Error Distributions

Each of the three variance specifications above is estimated under two conditional error distributions, yielding six models in total.

Under the **Normal** (Gaussian) distribution, the standardized innovations  $z_t = \varepsilon_t/\sigma_t$  are assumed i.i.d.  $\mathcal{N}(0, 1)$ .

Under the **Student- $t$**  distribution [14],  $z_t \sim t(\nu)$ , where the degrees-of-freedom parameter  $\nu > 2$  is estimated from the data jointly with the variance equation parameters. A smaller  $\nu$  implies heavier tails: the distribution assigns higher probability to extreme return realizations relative to the normal. Values of  $\nu < 10$  are generally taken as evidence of economically meaningful excess tail risk that the Gaussian assumption would understate. As documented in Section III, the excess kurtosis of 13.21 in the return data provides strong prior evidence that  $\nu$  will be estimated well below 10, and that the Student- $t$  models should produce materially better-calibrated VaR estimates in the tails.

All six GARCH models are estimated by Maximum Likelihood on the training set using the `arch` Python library. Model selection is based on the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC), both of which penalize log-likelihood for the number of estimated parameters:

$$\text{AIC} = -2 \ln \hat{L} + 2k, \quad \text{BIC} = -2 \ln \hat{L} + k \ln T, \quad (8)$$

where  $\hat{L}$  is the maximized likelihood,  $k$  is the number of free parameters, and  $T$  is the sample size. Lower values of AIC and BIC indicate a better balance between fit and parsimony.

## E. HAR-RV Model

The Heterogeneous Autoregressive model of Realized Volatility (HAR-RV) of Corsi [7] serves as the semi-parametric benchmark. Since intraday transaction data are not available for this study, realized volatility is proxied by the absolute daily log return  $\text{RV}_t = |r_t|$ , following the approach of Ding, Granger, and Engle [16].

The multi-horizon components are constructed as:

$$RV_t^{(d)} = |r_t|, \quad (9)$$

$$RV_t^{(w)} = \frac{1}{5} \sum_{i=1}^5 |r_{t-i+1}|, \quad (10)$$

$$RV_t^{(m)} = \frac{1}{22} \sum_{i=1}^{22} |r_{t-i+1}|, \quad (11)$$

representing the daily, weekly (five-day), and monthly (twenty-two-day) realized volatility averages, respectively.

The HAR-RV regression is then:

$$RV_{t+1} = c + \beta_d RV_t^{(d)} + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + \varepsilon_{t+1}, \quad (12)$$

estimated by Ordinary Least Squares (OLS). The economic interpretation of the three slope coefficients is direct. The daily coefficient  $\beta_d$  reflects the persistence of short-run shocks generated by high-frequency traders. The weekly coefficient  $\beta_w$  captures medium-horizon momentum from speculative activity. The monthly coefficient  $\beta_m$  embeds the slow-moving risk perceptions of institutional and long-term investors. Despite the simplicity of OLS estimation, the model generates long-memory-like autocorrelation in its volatility forecasts through this cascade of heterogeneous components, which is the key mechanism behind its competitive forecasting performance [7].

#### F. Expanding-Window Out-of-Sample Forecasting

All out-of-sample volatility forecasts are produced using an *expanding-window* scheme. At each step  $t$  in the test period, the model is re-estimated on all available data from the beginning of the sample up to and including day  $t$ , and a one-step-ahead forecast  $\hat{\sigma}_{t+1}$  (or  $\hat{RV}_{t+1}$  for HAR-RV) is produced for day  $t + 1$ . The estimation window then expands by one observation, and the process repeats for every test day. This procedure is illustrated schematically as follows:

- Step 1: Fit on  $[1, T_0] \rightarrow$  Forecast  $\hat{\sigma}_{T_0+1}$
- Step 2: Fit on  $[1, T_0 + 1] \rightarrow$  Forecast  $\hat{\sigma}_{T_0+2}$
- $\vdots$
- Step  $n$ : Fit on  $[1, T_0 + n - 1] \rightarrow$  Forecast  $\hat{\sigma}_{T_0+n}$

where  $T_0 = 3,017$  is the initial training size.

The expanding-window scheme is chosen over the alternative rolling-window approach for two reasons. First, it exploits the maximum amount of historical information at every forecast origin, which is generally more efficient when the data-generating process is stable. Second, it mirrors the approach a practitioner would follow in a real-time risk management setting, where all historical data are retained and used to continuously update model estimates. Crucially, the *identical* methodology is applied to both the GARCH family and the HAR-RV benchmark, ensuring that the comparison is not contaminated by differences in the estimation scheme.

Forecast accuracy in the out-of-sample period is evaluated using three standard loss functions. Let  $\hat{\sigma}_t$  denote the volatility forecast and  $RV_t = |r_t|$  denote the realized volatility proxy:

$$MSE = \frac{1}{T} \sum_{t=1}^T (\hat{\sigma}_t - RV_t)^2, \quad (13)$$

$$MAE = \frac{1}{T} \sum_{t=1}^T |\hat{\sigma}_t - RV_t|, \quad (14)$$

$$QLIKE = \frac{1}{T} \sum_{t=1}^T \left( \ln \hat{\sigma}_t^2 + \frac{RV_t^2}{\hat{\sigma}_t^2} \right). \quad (15)$$

MSE penalizes large forecast errors disproportionately and is therefore sensitive to outliers during stress episodes. MAE treats all errors equally and provides a more robust measure of average forecast deviation. QLIKE is an asymmetric loss function derived from the Gaussian quasi-likelihood: it penalizes under-prediction of volatility more heavily than over-prediction, making it particularly relevant for risk management applications where the cost of underestimating risk exceeds the cost of overestimating it.

## G. Value-at-Risk and Expected Shortfall

### 1. Parametric VaR and ES Formulas

Under the parametric approach, the one-day-ahead VaR at confidence level  $1 - \alpha$  is computed directly from the conditional volatility forecast:

$$\text{VaR}_{t+1}(\alpha) = \hat{\mu}_{t+1} + F_z^{-1}(\alpha) \cdot \hat{\sigma}_{t+1}, \quad (16)$$

where  $\hat{\mu}_{t+1}$  is the one-step-ahead conditional mean from the ARIMA equation (set to zero for HAR-RV),  $\hat{\sigma}_{t+1}$  is the conditional volatility forecast, and  $F_z^{-1}(\alpha)$  is the  $\alpha$ -quantile of the assumed standardized residual distribution. For the Normal distribution,  $F_z^{-1}(\alpha) = \Phi^{-1}(\alpha)$ ; for the Student- $t$  with  $\hat{\nu}$  degrees of freedom,  $F_z^{-1}(\alpha) = t_{\hat{\nu}}^{-1}(\alpha)$ . Since  $\alpha < 0.5$ , both quantiles are negative, and VaR is reported as a negative number representing the lower bound of the return distribution.

The Expected Shortfall at confidence level  $1 - \alpha$  is computed as:

$$\text{ES}_{t+1}(\alpha) = \hat{\mu}_{t+1} - \frac{f_z(F_z^{-1}(\alpha))}{\alpha} \cdot \hat{\sigma}_{t+1}, \quad (17)$$

where  $f_z(\cdot)$  is the probability density function of the standardized residual distribution. For the Student- $t$ , the ES expression becomes:

$$\text{ES}_{t+1}(\alpha) = \hat{\mu}_{t+1} - \frac{f_t(t_{\hat{\nu}}^{-1}(\alpha); \hat{\nu}) \cdot \left( \hat{\nu} + [t_{\hat{\nu}}^{-1}(\alpha)]^2 \right)}{\alpha (\hat{\nu} - 1)} \cdot \hat{\sigma}_{t+1}, \quad (18)$$

where  $f_t(\cdot; \hat{\nu})$  is the Student- $t$  density. ES is always more negative than VaR, reflecting the fact that it measures the average loss conditional on the loss exceeding the VaR threshold, and therefore always provides a more conservative risk estimate. VaR and ES are computed at both the 95% and 99% confidence levels.

### 2. HAR-RV-Based VaR

For the HAR-RV model, which does not directly produce a parametric distribution, the volatility forecast  $\widehat{\text{RV}}_{t+1}$  is used as the conditional standard deviation  $\hat{\sigma}_{t+1}$  in equations (16)–(17), with the Normal distribution assumed for the standardized residuals. This approach is consistent with the semi-parametric interpretation of the HAR-RV and allows its VaR estimates to be evaluated on a directly comparable basis with those of the GARCH-Normal models.

## H. Backtesting Procedures

### 1. Kupiec Proportion of Failures Test

The Kupiec [11] Proportion of Failures (POF) test assesses whether the empirical VaR violation rate is statistically consistent with the nominal confidence level. A violation on day  $t$  is defined as:

$$I_t = \mathbf{1}[r_t < \text{VaR}_t(\alpha)], \quad (19)$$

where  $I_t = 1$  if the realized return falls below the VaR threshold (a violation) and  $I_t = 0$  otherwise. Let  $N = \sum_t I_t$  be the total number of violations over  $T$  test observations, and  $\hat{p} = N/T$  the empirical violation rate. The null hypothesis is  $H_0 : \hat{p} = \alpha$  (the observed violation rate equals the nominal level). The likelihood ratio test statistic is:

$$\text{LR}_{\text{uc}} = -2 \ln \frac{(1 - \alpha)^{T-N} \alpha^N}{(1 - \hat{p})^{T-N} \hat{p}^N} \xrightarrow{d} \chi^2(1), \quad (20)$$

under  $H_0$ . A model with too many violations ( $\hat{p} > \alpha$ ) underestimates risk; one with too few ( $\hat{p} < \alpha$ ) is over-conservative. Both cases fail the test when  $p$ -value  $< 0.05$ .

## 2. Christoffersen Conditional Coverage Test

The Christoffersen [12] Conditional Coverage (CC) test jointly evaluates unconditional coverage and the independence of violations. A model may produce the correct average violation rate but still be misspecified if violations cluster in time (for instance, occurring on consecutive trading days during a stress episode), which would indicate that the model fails to adapt quickly enough to changing volatility regimes.

The test models the violation indicator sequence  $\{I_t\}$  as a first-order Markov chain, estimating the transition probabilities:

$$\hat{\pi}_{ij} = \frac{n_{ij}}{\sum_j n_{ij}}, \quad i, j \in \{0, 1\}, \quad (21)$$

where  $n_{ij}$  is the number of transitions from state  $i$  to state  $j$ . Under the null hypothesis of independence,  $\pi_{01} = \pi_{11}$ , meaning the probability of a violation is the same regardless of whether a violation occurred on the previous day. The likelihood ratio statistic for independence is:

$$\text{LR}_{\text{ind}} = -2 \ln \frac{(1 - \hat{\pi})^{n_{00} + n_{10}} \hat{\pi}^{n_{01} + n_{11}}}{(1 - \hat{\pi}_{01})^{n_{00}} \hat{\pi}_{01}^{n_{01}} (1 - \hat{\pi}_{11})^{n_{10}} \hat{\pi}_{11}^{n_{11}}} \xrightarrow{d} \chi^2(1), \quad (22)$$

where  $\hat{\pi} = (n_{01} + n_{11}) / (n_{00} + n_{01} + n_{10} + n_{11})$  is the unconditional violation rate. The joint Conditional Coverage statistic is:

$$\text{LR}_{\text{cc}} = \text{LR}_{\text{uc}} + \text{LR}_{\text{ind}} \xrightarrow{d} \chi^2(2). \quad (23)$$

A well-specified VaR model must pass both the Kupiec and Christoffersen tests at both the 95% and 99% confidence levels. Passing the Kupiec test alone is insufficient: a model that generates the correct average number of violations but clusters them during periods of acute market stress – such as the COVID-19 crash or the 2022 bear market – provides a misleading picture of risk and fails to satisfy the independence requirement essential for regulatory compliance under Basel III [10].

## V. Empirical Results

This section presents and interprets the empirical findings across all dimensions of the analysis, proceeding in the order of the modelling pipeline established in Section IV. The discussion covers stationarity and ARIMA diagnostics, GARCH in-sample estimation and model selection, HAR-RV estimation, out-of-sample forecast accuracy, VaR and ES calibration, and formal backtesting.

### A. Stationarity Tests and ARIMA Results

#### 1. Stationarity Tests

Table 4 reports the ADF and KPSS test results for the raw closing-price series and the log-return series.

**Table 4 Stationarity Test Results for S&P 500 Closing Price and Log Returns. ADF null hypothesis: unit root (non-stationary); reject  $H_0$  implies stationary. KPSS null hypothesis: stationary; failure to reject  $H_0$  implies stationary.**

| Series        | ADF Stat | ADF $p$ -value | KPSS Stat | Conclusion     |
|---------------|----------|----------------|-----------|----------------|
| Closing price | 1.4323   | 0.9973         | 9.0598    | Non-stationary |
| Log return    | -13.3438 | < 0.001        | 0.0274    | Stationary     |

The raw closing-price series is clearly non-stationary: the ADF statistic of 1.43 fails to reject the unit-root null ( $p = 0.997$ ), and the KPSS statistic of 9.06 strongly rejects the stationarity null. This is the expected outcome for an equity price index, which follows an upward trend over time. The log-return series, by contrast, is unambiguously

stationary: the ADF statistic of  $-13.34$  ( $p < 0.001$ ) decisively rejects the unit-root null, and the KPSS statistic of  $0.027$  fails to reject stationarity ( $p \geq 0.10$ ). Both tests thus agree, providing the strongest possible joint evidence that log returns are  $I(0)$  stationary, as required for valid ARIMA and GARCH estimation.

## 2. ARIMA Mean Equation

The `auto_arima` stepwise search over AIC selected **ARIMA(4, 0, 0)** as the optimal mean specification (AIC = 8,972.27; BIC = 9,008.34), representing an improvement of 12.35 points over the next-best specification, ARIMA(1, 0, 1), and 90.58 points over a constant mean equation. All four AR coefficients are statistically significant at the 5% level, with estimates  $\hat{\phi}_1 = -0.137$  ( $p < 0.001$ ),  $\hat{\phi}_2 = +0.083$  ( $p < 0.001$ ),  $\hat{\phi}_3 = -0.018$  ( $p = 0.040$ ), and  $\hat{\phi}_4 = -0.061$  ( $p < 0.001$ ). The alternating signs indicate a mild short-run mean-reversion pattern, consistent with bid–ask bounce and short-horizon return reversal documented in the equity microstructure literature. Although the magnitudes are modest, extracting this predictable mean structure ensures that GARCH models are applied to the pure innovation sequence rather than to autocorrelation-contaminated returns. As a robustness check, the complete estimation and backtesting pipeline was re-estimated under a constant mean equation; all pass/fail conclusions in Table 17 remain qualitatively unchanged, confirming that the choice of mean specification does not drive the main findings (Table 23).

## 3. ARCH Effect Test

**Table 5** ARCH-LM Test Results for ARIMA(4, 0, 0) Residuals (10 lags).  $H_0$ : no ARCH effect (constant variance). Rejection confirms conditional heteroskedasticity and justifies GARCH modelling.

| Test                 | Statistic | $p$ -value | Conclusion            |
|----------------------|-----------|------------|-----------------------|
| ARCH-LM ( $\chi^2$ ) | 982.20    | $< 0.001$  | ARCH effect confirmed |
| $F$ -statistic       | 145.33    | $< 0.001$  | ARCH effect confirmed |

The ARCH-LM test results are reported in Table 5. The  $\chi^2$  statistic of 982.20 ( $p < 0.001$ ) overwhelmingly rejects the null hypothesis of no ARCH effect. This finding confirms that the variance of the ARIMA residuals is far from constant over time and that a GARCH-type model is statistically warranted. The visual evidence presented in Section IV (Figures 5–7) corroborates this conclusion: squared residuals exhibit strongly persistent autocorrelation at all lags up to 30, with autocorrelations at short lags exceeding 0.50.

## B. GARCH Model Estimation

### 1. In-Sample Fit and Model Selection

Table 6 reports the estimated parameters and information criteria for all six GARCH-family models, estimated by Maximum Likelihood on the training set ( $T = 3,017$  observations).

Figure 8 displays the AIC and BIC values for all six models as horizontal bar charts, making the ranking immediately visible. Both criteria agree unambiguously: the three Student- $t$  models occupy the bottom of the ranking (lowest AIC/BIC), and EGARCH(1,1)-Student- $t$  is the clear winner.

The results reveal a clear pattern: Student- $t$  specifications uniformly dominate their Gaussian counterparts within each model class. Switching from Normal to Student- $t$  reduces the AIC by 210.5 points for GARCH(1,1), by 230.2 points for GJR-GARCH(1,1), and by 218.0 points for EGARCH(1,1), confirming the strong empirical relevance of fat-tailed return innovations in S&P 500 daily returns. Within the Student- $t$  family, asymmetric specifications further outperform the symmetric baseline, with AIC improvements of 117.9 and 129.7 points for GJR-GARCH(1,1) and EGARCH(1,1) respectively, providing formal statistical support for the leverage effect hypothesis.

The boundary solution  $\hat{\alpha} = 0.000$  obtained for GJR-GARCH(1,1)-Student- $t$  warrants further examination. A restricted specification imposing  $\alpha \equiv 0$  yields a log-likelihood of  $-3,621.66$  against  $-3,621.62$  for the unrestricted model, producing a likelihood ratio statistic of  $0.078$  ( $p = 0.780$ ,  $\chi^2(1)$ ) that fails to reject the null at any conventional significance level. The restricted model additionally achieves lower AIC (7,251.33 vs. 7,253.25) and BIC (7,275.37 vs. 7,283.31), while the leverage coefficient remains stable across specifications ( $\hat{\gamma} = 0.303$  vs. 0.302). These results confirm that the boundary solution does not constitute an identification failure: the symmetric ARCH component carries no independent information once the asymmetric leverage effect is accounted for (Table 24).



**Table 6 GARCH-Family Model Estimation Results (Training Set, January 2010 – December 2021). Parameters ( $\omega, \alpha, \beta, \gamma, \nu$ ) are the variance intercept, ARCH coefficient, GARCH coefficient, asymmetry parameter, and Student- $t$  degrees of freedom, respectively. Persistence =  $\alpha + \beta$  for GARCH/GJR; reported as  $|\beta|$  from the log-variance equation for EGARCH. The best model by AIC and BIC is EGARCH(1,1)-Student- $t$ .**

| Model              | Dist.                         | $\omega$                   | $\alpha$         | $\beta$          | $\gamma$          | $\nu$           | AIC            | BIC            | Persist.     |
|--------------------|-------------------------------|----------------------------|------------------|------------------|-------------------|-----------------|----------------|----------------|--------------|
| GARCH(1,1)         | Normal                        | 0.0421***                  | 0.1887***        | 0.7745***        | –                 | –               | 7531.25        | 7555.30        | 0.963        |
| GARCH(1,1)         | Student- $t$                  | 0.0293***                  | 0.1924***        | 0.7973***        | –                 | 4.999***        | 7320.70        | 7350.76        | 0.990        |
| GJR-GARCH(1,1)     | Normal                        | 0.0429***                  | 0.0439           | 0.7854***        | 0.2590***         | –               | 7432.96        | 7463.02        | 0.829        |
| GJR-GARCH(1,1)     | Student- $t$                  | 0.0330***                  | 0.0000           | 0.8066***        | 0.3353***         | 5.323***        | 7202.80        | 7238.87        | 0.807        |
| EGARCH(1,1)        | Normal                        | –0.0091                    | 0.2484***        | 0.9426***        | –0.1743***        | –               | 7408.99        | 7439.05        | 1.191        |
| <b>EGARCH(1,1)</b> | <b>Student-<math>t</math></b> | <b>–0.0118<sup>†</sup></b> | <b>0.2047***</b> | <b>0.9594***</b> | <b>–0.2107***</b> | <b>5.491***</b> | <b>7191.03</b> | <b>7227.10</b> | <b>1.164</b> |

\*\*\* significant at 1%; <sup>†</sup>  $p = 0.063$  (marginally insignificant at 5%).

Persistence =  $\alpha + \beta$  for GARCH and GJR-GARCH; Persistence =  $|\beta|$  for EGARCH (values above 1 are permissible).

GJR-GARCH(1,1)-Student- $t$ :  $\alpha$  estimate is effectively zero (numerical boundary); leverage effect fully captured by  $\gamma = 0.3353$ \*\*\*.

The best model by both AIC (7,191.03) and BIC (7,227.10) is the **EGARCH(1,1)-Student- $t$** . Its estimated parameters carry the following economic interpretations. The ARCH coefficient  $\hat{\alpha} = 0.205$  ( $p < 0.001$ ) indicates that the standardized magnitude of yesterday's shock has a substantial immediate effect on today's variance. The GARCH coefficient  $\hat{\beta} = 0.959$  ( $p < 0.001$ ) reflects very high persistence: a volatility shock decays by only 4.1% per day, implying that elevated volatility takes approximately three to four months to revert to its long-run mean. The asymmetry parameter  $\hat{\gamma} = -0.211$  ( $p < 0.001$ ) is negative and highly significant, confirming the leverage effect: negative return shocks increase log-variance by (–0.211) additional units compared with positive shocks of equal magnitude, consistent with the well-documented finding that bad news amplifies equity market volatility more than good news. The estimated degrees-of-freedom parameter  $\hat{\nu} = 5.49$  ( $p < 0.001$ ) is well below 10, confirming economically meaningful fat tails that could not be captured by the Gaussian assumption. This value implies that the conditional return distribution assigns substantially more probability to extreme losses than a normal distribution, a property that is directly relevant to the accuracy of VaR estimates at high confidence levels.

Figure 9 displays the conditional volatility estimates from the best model (EGARCH(1,1)-Student- $t$ ) over the full sample, with key market stress episodes highlighted. The model captures the sharp COVID-19 spike to  $\approx 2.6\%$  in March 2020 and the elevated but more gradual volatility regime of 2022, demonstrating its ability to respond to both acute and persistent stress episodes. The leverage effect parameter  $\hat{\gamma} = -0.211$  enables the model to react more sharply to negative shocks than to positive ones of equal magnitude, which is clearly visible in the asymmetric spikes around the two stress periods.

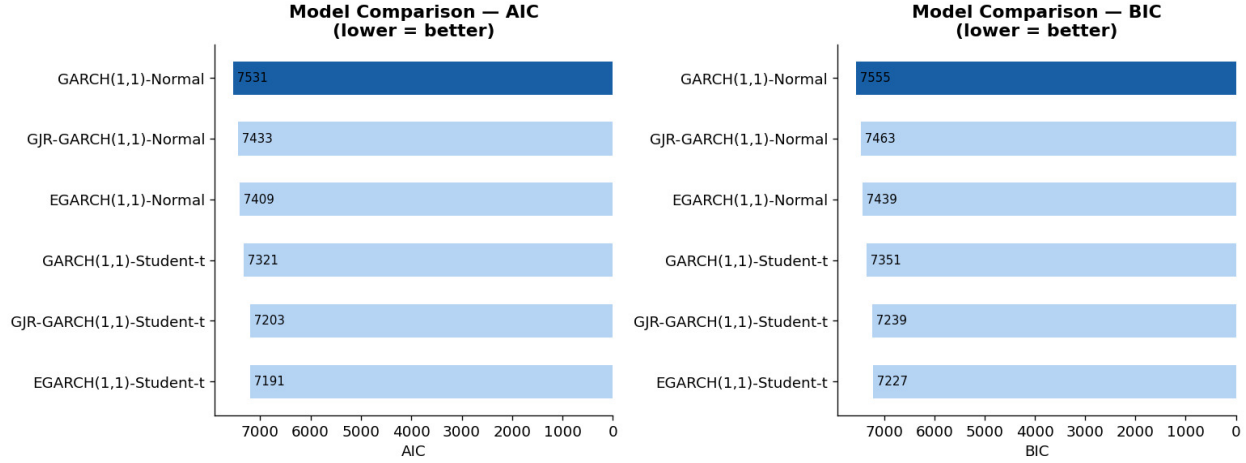
### C. HAR-RV Model Results

Table 7 reports the OLS estimates for the HAR-RV model estimated on the training set.

**Table 7 HAR-RV OLS Regression Results (Training Set, January 2010 – December 2021,  $N = 2,995$  observations after lag construction). Dependent variable:  $RV_{t+1} = |r_{t+1}|$ . Standard errors are heteroskedasticity-robust.**

| Component           | Coefficient           | Std. Error | $t$ -stat | $p$ -value |
|---------------------|-----------------------|------------|-----------|------------|
| Intercept ( $c$ )   | 0.1377                | 0.0225     | 6.11      | < 0.001    |
| $\beta_d$ (Daily)   | –0.0105               | 0.0221     | –0.48     | 0.635      |
| $\beta_w$ (Weekly)  | 0.6201                | 0.0415     | 14.96     | < 0.001    |
| $\beta_m$ (Monthly) | 0.1922                | 0.0415     | 4.63      | < 0.001    |
| $R^2$               | 0.281                 |            |           |            |
| $F$ -statistic      | 390.2 ( $p < 0.001$ ) |            |           |            |

Several findings are noteworthy. First, the overall model is highly significant ( $F = 390.2$ ,  $p < 0.001$ ), confirming that the multi-horizon structure of the HAR-RV captures genuine predictive information about future realized volatility.



**Fig. 8 GARCH-Family Model Comparison by AIC and BIC (Training Set, January 2010 – December 2021).** *Left panel:* AIC values (lower = better in-sample fit penalised for parameters). *Right panel:* BIC values (stricter penalty for additional parameters). Both criteria rank EGARCH(1,1)-Student- $t$  first (AIC = 7,191; BIC = 7,227) and GARCH(1,1)-Normal last (AIC = 7,531; BIC = 7,555). The consistent ranking across both criteria confirms that the fat-tailed and asymmetric specifications are not merely over-parameterised fits.

The  $R^2 = 0.281$  indicates that 28.1% of next-day absolute return variation is explained by its own history at three horizons, which is a substantial proportion for a one-step-ahead forecast of a noisy daily series. Second, the dominant forecasting contribution comes from the **weekly component**, with  $\hat{\beta}_w = 0.621$  ( $p < 0.001$ ). This coefficient implies that a one-unit increase in average weekly volatility is associated with a 0.621 increase in tomorrow’s realized volatility, holding the other components constant. The large and significant weekly coefficient supports the Heterogeneous Market Hypothesis: the primary volatility signal comes from participants with a multi-day horizon (speculative traders and short-term fund managers), not from intraday noise.

Third, the **monthly component** is also significant ( $\hat{\beta}_m = 0.192$ ,  $p < 0.001$ ), capturing the slow-moving volatility expectations of institutional investors. Together,  $\hat{\beta}_w + \hat{\beta}_m = 0.812$ , implying strong medium-to-long-horizon persistence in the volatility process.

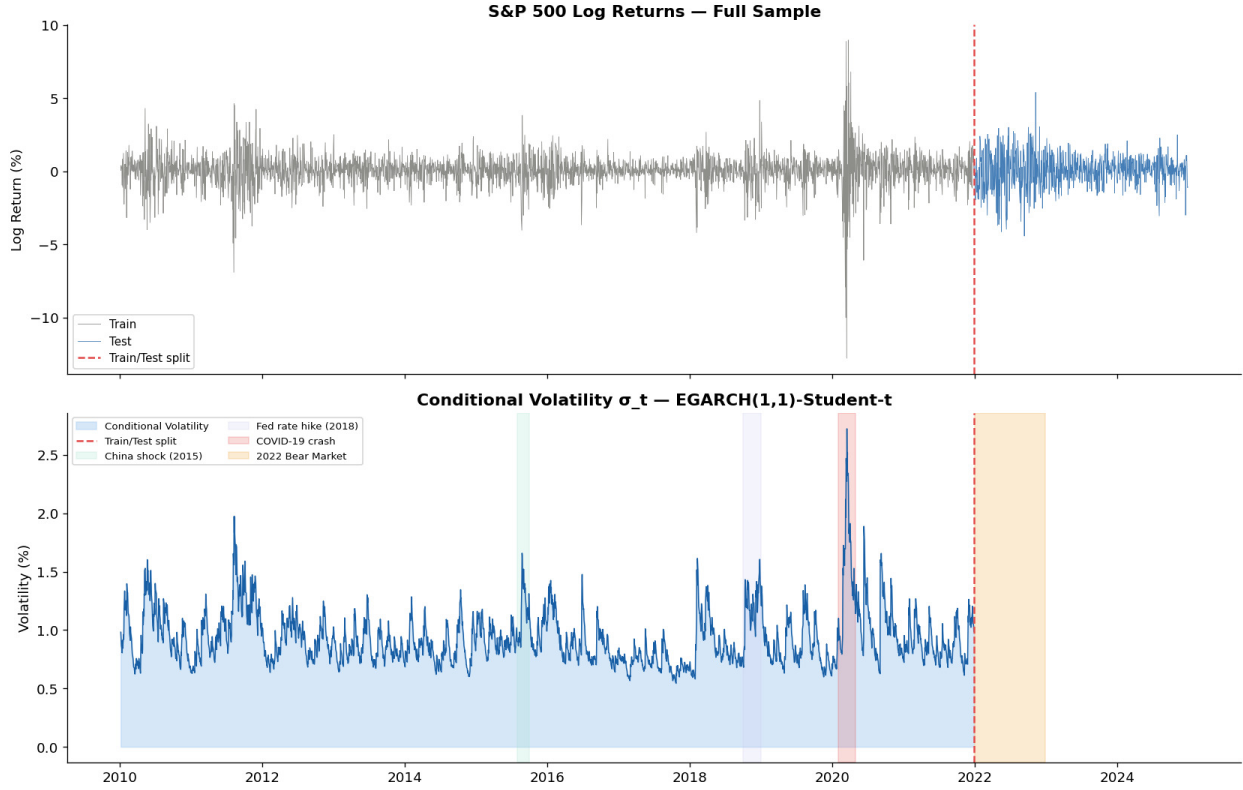
Fourth, the **daily component**  $\hat{\beta}_d = -0.010$  is not statistically significant ( $p = 0.635$ ). This finding suggests that single-day idiosyncratic noise in  $|r_t|$  has no independent predictive value once the weekly and monthly averages are controlled for. The practical implication is that, when using absolute returns as a realized volatility proxy, the HAR-RV effectively reduces to a two-component (weekly and monthly) specification.

#### D. Out-of-Sample Volatility Forecast Accuracy

Table 8 reports the out-of-sample forecast accuracy metrics for all seven models over the 755-day test period (December 2021 – December 2024), produced using the expanding-window scheme described in Section IV.F. Figure 10 provides a visual overview of the forecasted conditional volatility paths from all six GARCH models over the test period.

The results present a nuanced picture. Under both MSE and MAE, **HAR-RV achieves the best out-of-sample forecast performance** across all seven models, with MAE = 0.542 and MSE = 0.504. The best-performing GARCH model under MAE is EGARCH(1,1)-Normal (MAE = 0.586), representing a 8.1% deterioration relative to HAR-RV. The pattern is consistent across all GARCH specifications: Student- $t$  models are systematically outperformed by their Normal counterparts under both MSE and MAE, suggesting that the distributional assumption has limited impact on the point forecast of volatility but plays a more important role in tail risk quantification, as confirmed in the backtesting results below.

Under the QLIKE loss function, however, the ranking reverses: **EGARCH(1,1)-Normal achieves the lowest QLIKE** (= 0.988), followed closely by EGARCH(1,1)-Student- $t$  (= 0.991), while HAR-RV produces the worst QLIKE (= 1.306) among all seven models. This divergence between MAE/MSE and QLIKE rankings is economically meaningful. QLIKE is an asymmetric loss function that penalizes under-prediction of variance more heavily than over-prediction, which makes it particularly informative for risk management applications. The poor QLIKE performance of HAR-RV,



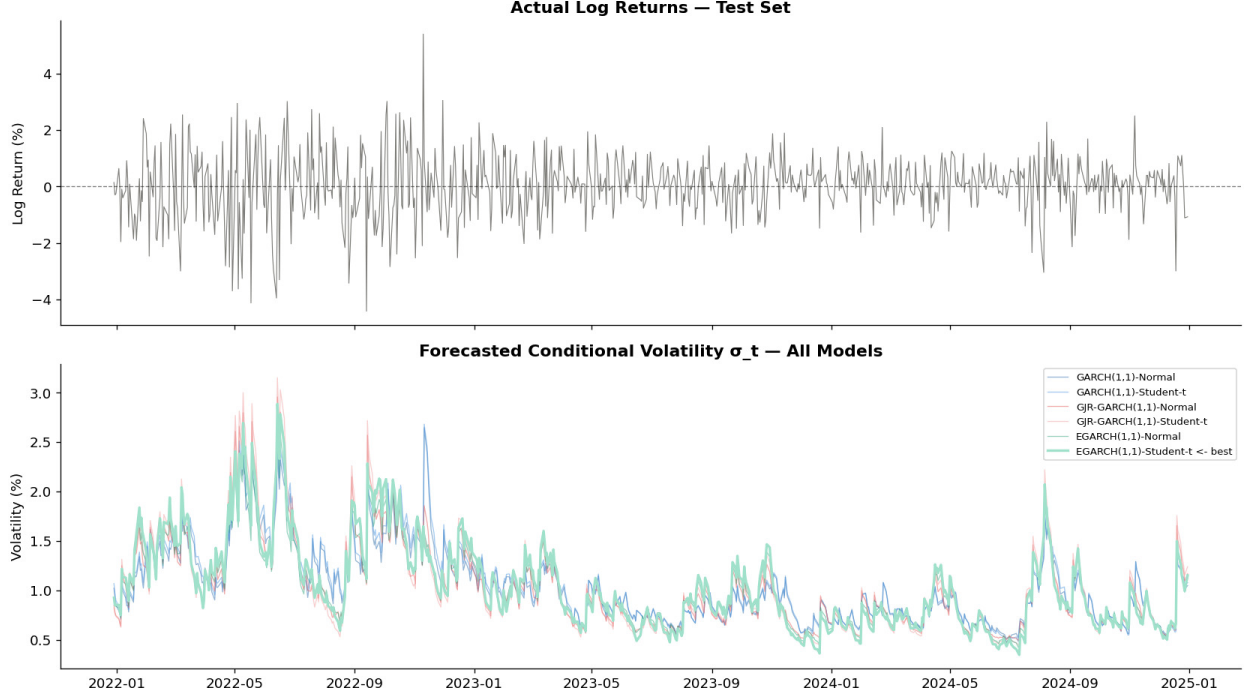
**Fig. 9** S&P 500 Log Returns and Conditional Volatility  $\hat{\sigma}_t$  from EGARCH(1,1)-Student- $t$  (Full Sample, January 2010 – December 2024). *Upper panel:* Daily log returns; grey = training set, blue = test set; red dashed line marks the train/test split (December 2021). *Lower panel:* Estimated conditional volatility  $\hat{\sigma}_t$  (%) with shading for the China shock (2015, green), Fed rate-hiking cycle (2018, blue), COVID-19 crash (2020, pink), and 2022 bear market (orange). The model clearly resolves all four stress regimes as distinct volatility elevations against an otherwise calm baseline.

despite its leading MAE and MSE, suggests that its volatility forecasts are too low in periods of elevated risk, which is precisely when accurate calibration matters most. This finding is partly attributable to the use of  $|r_t|$  as a realized volatility proxy: since  $|r_t|$  is a noisy one-observation estimator that does not account for intraday variation, the HAR-RV can only capture average-magnitude volatility and may systematically understate extreme tail episodes.

Table 19 extends the Diebold-Mariano analysis to the QLIKE loss function, which penalises volatility under-prediction asymmetrically and is therefore the most operationally relevant criterion for risk management applications. The results reveal a complete reversal of the forecast ranking. Under QLIKE, all six GARCH specifications significantly outperform HAR-RV at the 0.1% level, with DM statistics ranging from 5.37 to 5.83. Notably, these test statistics are larger in absolute magnitude than those obtained under MSE and MAE, indicating that the GARCH advantage in tail-region calibration is statistically more decisive than the HAR-RV advantage in point accuracy.

The simultaneous significance in opposite directions across loss functions constitutes formal statistical evidence of a fundamental trade-off between the two modelling paradigms. Specifically, HAR-RV wins under symmetric loss (MSE and MAE), whereas GARCH wins under asymmetric loss (QLIKE). While HAR-RV achieves lower average forecast error across all trading days, it systematically underestimates volatility precisely during elevated-risk periods, which carry the highest weight under QLIKE. This duality explains why HAR-RV's point-forecast superiority does not translate into regulatory VaR validity: a model that performs well on typical low-volatility days but fails on the tail-risk days that matter most for capital adequacy will satisfy symmetric accuracy criteria while failing backtesting requirements under Basel III.

The EGARCH models rank best among all GARCH specifications across all three loss functions, consistent with the leverage-effect framework: the asymmetric response to negative shocks enables the EGARCH to react more rapidly to



**Fig. 10** Forecasted Conditional Volatility  $\hat{\sigma}_t$  from All GARCH Models (Test Set, December 2021 – December 2024,  $T = 755$  observations). *Upper panel:* Actual daily log returns. *Lower panel:* One-step-ahead volatility forecasts from all six GARCH specifications. The EGARCH(1,1)-Student- $t$  (best by AIC, highlighted in green) produces the most elevated forecasts during the 2022 bear market, while all models converge to similar low-volatility estimates during the calmer 2023–2024 recovery. The spread between models is largest during the stress period of early-to-mid 2022, where leverage-asymmetric models (GJR, EGARCH) diverge upward from the symmetric GARCH(1,1).

downside market events and thereby produce more accurate tail-region forecasts.

Figure 11 provides a three-panel visual decomposition of the HAR-RV versus best-GARCH comparison across the test period.

## E. Value-at-Risk and Expected Shortfall

### 1. Average Risk Estimates

Table 9 reports the average VaR and ES estimates over the test period for all seven models at the 95% and 99% confidence levels.

The comparison between the best GARCH model and HAR-RV reveals a striking difference in the implied risk levels. The EGARCH(1,1)-Student- $t$  produces an average VaR 99% of  $-3.52\%$ , which is approximately **91% more conservative** than the HAR-RV's average VaR 99% of  $-1.85\%$ . Similarly, the EGARCH's average ES 99% of  $-4.66\%$  is more than twice as large as HAR-RV's  $-2.12\%$ . This divergence has a straightforward explanation: the GARCH models scale VaR and ES by the conditional standard deviation and apply the quantile of the Student- $t$  distribution, which has heavy tails. By contrast, the HAR-RV uses  $|r_t|$  as its volatility estimate, which averages out short-term extremes, and applies Normal quantiles, making it structurally less sensitive to tail events.

From a regulatory perspective, the ES is the more informative metric: ES captures not just whether a loss exceeds the VaR threshold but also how severe that loss is on average when it does. The EGARCH(1,1)-Student- $t$ 's ES 99% of  $-4.66\%$  indicates that on the 1% worst days, the expected loss exceeds 4.66% on average, compared with only 2.12% under the HAR-RV. Under the Basel III framework [10], an institution using the HAR-RV-based ES would hold substantially less capital than one using the EGARCH-Student- $t$ , which may be inadequate during stress episodes of the kind observed in March 2020 or during the sustained drawdown of 2022.

**Table 8 Out-of-Sample Forecast Accuracy: MSE, MAE, and QLIKE (Test Set, December 2021 – December 2024,  $T = 755$  observations). Lower values indicate better forecast performance. Bold entries indicate the best value for each metric. Realized volatility proxy:  $RV_t = |r_t|$ .**

| Model          | Dist.        | MSE           | MAE           | QLIKE         |
|----------------|--------------|---------------|---------------|---------------|
| <b>HAR-RV</b>  | –            | <b>0.5041</b> | <b>0.5421</b> | 1.3059        |
| EGARCH(1,1)    | Normal       | 0.5503        | 0.5857        | <b>0.9881</b> |
| EGARCH(1,1)    | Student- $t$ | 0.5679        | 0.5887        | 0.9912        |
| GJR-GARCH(1,1) | Normal       | 0.5734        | 0.5910        | 1.0053        |
| GJR-GARCH(1,1) | Student- $t$ | 0.5994        | 0.5965        | 1.0057        |
| GARCH(1,1)     | Normal       | 0.5743        | 0.6054        | 1.0283        |
| GARCH(1,1)     | Student- $t$ | 0.5936        | 0.6143        | 1.0244        |

**Table 9 Average Daily VaR and ES on the Test Set (%), December 2021 – December 2024 ( $T = 755$  observations). Negative values represent expected losses (lower bounds of the return distribution). More negative = more conservative estimate. VaR and ES are computed parametrically using the expanding-window conditional volatility forecast.**

| Model          | Dist.        | Avg VaR 95% | Avg VaR 99% | Avg ES 99% |
|----------------|--------------|-------------|-------------|------------|
| GARCH(1,1)     | Normal       | –1.70%      | –2.41%      | –2.76%     |
| GARCH(1,1)     | Student- $t$ | –2.15%      | –3.59%      | –4.75%     |
| GJR-GARCH(1,1) | Normal       | –1.68%      | –2.38%      | –2.73%     |
| GJR-GARCH(1,1) | Student- $t$ | –2.11%      | –3.52%      | –4.65%     |
| EGARCH(1,1)    | Normal       | –1.69%      | –2.39%      | –2.73%     |
| EGARCH(1,1)    | Student- $t$ | –2.11%      | –3.52%      | –4.66%     |
| HAR-RV         | Normal       | –1.31%      | –1.85%      | –2.12%     |

Note: VaR and ES computed parametrically using expanding-window conditional volatility forecasts.

Student- $t$  specifications are systematically more conservative than Normal counterparts at equivalent volatility levels, owing to heavier quantiles at  $\hat{\nu} \approx 5.0$ – $5.5$  degrees of freedom. HAR-RV applies Normal quantiles to  $|r_t|$  as the volatility proxy.

Full parameter estimates for all models are reported in Appendix A, Table ??.

Figure 12 illustrates the VaR violation plot for the HAR-RV model, where the scale of the under-estimation problem is immediately apparent: the orange and red dashed VaR boundaries lie systematically close to zero, well above the magnitude of many actual loss realizations.

## 2. GARCH vs. Normal: The Role of Fat Tails

A consistent finding across all three GARCH specifications is that Student- $t$  models produce significantly more negative VaR and ES estimates than their Normal counterparts. Figure 13 makes this contrast visible across all six GARCH models simultaneously.

For GARCH(1,1), the Student- $t$  specification generates a VaR 99% that is approximately 30–40% more conservative than the Normal specification at equivalent conditional volatility levels, owing to the heavier quantile of the  $t$ -distribution with  $\hat{\nu} \approx 5.5$  degrees of freedom. This difference is directly consequential for risk management: a desk that uses a Normal GARCH model to compute capital requirements will systematically hold less capital than one using the Student- $t$  specification, and that shortfall will be most pronounced precisely in the tail events that capital is meant to cover.

## F. Backtesting Results

### 1. Kupiec and Christoffersen Tests

Table 10 reports the complete backtesting results for all seven models at both the 95% and 99% confidence levels.

**Table 10 Backtesting Results: Kupiec POF Test and Christoffersen Conditional Coverage Test (Test Set,  $T = 755$  observations). Expected violation rate: 5.00% at 95% CL, 1.00% at 99% CL. Pass criterion:  $p$ -value  $> 0.05$ . A well-specified model passes both tests at both confidence levels.**

| Model                                     | CL  | # Viol. | Exp. Viol. | Act. Viol. | Kupiec $p$ | Kupiec      | Christoff. $p$ | Christoff.  |
|-------------------------------------------|-----|---------|------------|------------|------------|-------------|----------------|-------------|
| GARCH(1,1)-Normal                         | 95% | 49      | 5.00%      | 6.49%      | 0.072      | Pass        | 0.450          | Pass        |
| GARCH(1,1)-Normal                         | 99% | 13      | 1.00%      | 1.72%      | 0.071      | Pass        | 0.499          | Pass        |
| GARCH(1,1)-Student- $t$                   | 95% | 23      | 5.00%      | 3.05%      | 0.008      | <b>Fail</b> | 0.229          | Pass        |
| GARCH(1,1)-Student- $t$                   | 99% | 3       | 1.00%      | 0.40%      | 0.058      | Pass        | 0.877          | Pass        |
| GJR-GARCH(1,1)-Normal                     | 95% | 55      | 5.00%      | 7.28%      | 0.007      | <b>Fail</b> | 0.571          | Pass        |
| GJR-GARCH(1,1)-Normal                     | 99% | 17      | 1.00%      | 2.25%      | 0.003      | <b>Fail</b> | 0.376          | Pass        |
| GJR-GARCH(1,1)-Student- $t$               | 95% | 24      | 5.00%      | 3.18%      | 0.014      | <b>Fail</b> | 0.789          | Pass        |
| GJR-GARCH(1,1)-Student- $t$               | 99% | 3       | 1.00%      | 0.40%      | 0.058      | Pass        | 0.877          | Pass        |
| EGARCH(1,1)-Normal                        | 95% | 50      | 5.00%      | 6.62%      | 0.051      | Pass        | 0.851          | Pass        |
| EGARCH(1,1)-Normal                        | 99% | 13      | 1.00%      | 1.72%      | 0.071      | Pass        | 0.499          | Pass        |
| <b>EGARCH(1,1)-Student-<math>t</math></b> | 95% | 23      | 5.00%      | 3.05%      | 0.008      | <b>Fail</b> | 0.229          | Pass        |
| <b>EGARCH(1,1)-Student-<math>t</math></b> | 99% | 4       | 1.00%      | 0.53%      | 0.154      | <b>Pass</b> | 0.836          | <b>Pass</b> |
| HAR-RV                                    | 95% | 83      | 5.00%      | 10.99%     | 0.000      | <b>Fail</b> | 0.959          | Pass        |
| HAR-RV                                    | 99% | 36      | 1.00%      | 4.77%      | 0.000      | <b>Fail</b> | 0.535          | Pass        |

### 2. Interpretation of Backtesting Results

The backtesting results reveal several important patterns.

**Normal models at 95% marginally adequate.** The GARCH(1,1)-Normal and EGARCH(1,1)-Normal both pass the Kupiec test at 95% with  $p$ -values of 0.072 and 0.051 respectively, although both are very close to the rejection threshold. Their violation rates of 6.49% and 6.62% are somewhat above the nominal 5.00%, indicating slight under-estimation of risk. Crucially, all models pass the Christoffersen independence test across all confidence levels and model specifications, indicating that violations are not temporally clustered. This is a positive finding: even during the 2022 bear market, no model generated runs of consecutive daily losses that exceeded its own VaR boundary, suggesting that the conditional variance dynamics update quickly enough to accommodate regime shifts.

**Student- $t$  models over-conservative at 95%.** The Student- $t$  specifications for GARCH(1,1) and GJR-GARCH(1,1) fail the Kupiec test at 95% with  $p$ -values of 0.008 and 0.014 respectively, because their violation rates of 3.05% and 3.18% are *too low* relative to the nominal 5.00%. Statistically, these models are over-conservative: they predict larger losses than actually materialise, which would lead to unnecessary capital reserves if used for regulatory purposes. This over-conservatism arises from the heavy tails of the Student- $t$  distribution with  $\hat{\nu} \approx 5.5$  degrees of freedom, which places very negative quantiles at the 5% level.

**EGARCH(1,1)-Student- $t$  best at 99% level.** The best in-sample model by AIC passes both the Kupiec ( $p = 0.154$ ) and Christoffersen ( $p = 0.836$ ) tests at the 99% confidence level, which is the level most relevant for regulatory capital under Basel III. Its 99% violation rate of 0.53% is below the nominal 1.00%, which means the model produces fewer losses than predicted, a conservative outcome that is appropriate for risk management purposes. At the 95% level, however, it fails the Kupiec test ( $p = 0.008$ ), exhibiting the same over-conservative pattern as the other Student- $t$  models.

**HAR-RV severely under-estimates tail risk.** The HAR-RV fails the Kupiec test at both confidence levels with  $p < 0.001$ , generating violation rates of 10.99% and 4.77% at the 95% and 99% levels respectively, compared with nominal rates of 5.00% and 1.00%. These violations are more than twice the expected frequency at both levels, indicating systematic and severe under-estimation of tail risk. The root cause is structural: the HAR-RV uses the average absolute return as a volatility proxy and Normal quantiles to compute VaR. This combination substantially underestimates the extreme quantiles of the return distribution, particularly during the 2022 bear market, where elevated

but less acute volatility persisted over several months and was not captured by a proxy anchored to recent absolute returns. Importantly, the HAR-RV passes the Christoffersen test at both levels ( $p = 0.959$  and  $p = 0.535$ ), confirming that while violations are far too frequent, they are not temporally clustered. This suggests that the HAR-RV's VaR failures are driven by a systematic level bias rather than a dynamic misspecification.

Figure 14 provides a visual summary of all Kupiec and Christoffersen  $p$ -values in heatmap form, making the pass/fail pattern immediately legible across all model-confidence-level combinations.

### 3. Overall Assessment

Table 11 summarizes the pass/fail outcomes across all models and confidence levels.

**Table 11 Backtesting Pass/Fail Summary. A model passes overall if it satisfies both Kupiec and Christoffersen tests at both confidence levels.**

| Model                       | Kupiec 95% | Kupiec 99% | Christoff. (all) | Overall |
|-----------------------------|------------|------------|------------------|---------|
| GARCH(1,1)-Normal           | Pass       | Pass       | Pass             | ✓       |
| GARCH(1,1)-Student- $t$     | Fail       | Pass       | Pass             | Partial |
| GJR-GARCH(1,1)-Normal       | Fail       | Fail       | Pass             | ×       |
| GJR-GARCH(1,1)-Student- $t$ | Fail       | Pass       | Pass             | Partial |
| EGARCH(1,1)-Normal          | Pass       | Pass       | Pass             | ✓       |
| EGARCH(1,1)-Student- $t$    | Fail       | Pass       | Pass             | Partial |
| HAR-RV                      | Fail       | Fail       | Pass             | ×       |

Two models, **GARCH(1,1)-Normal** and **EGARCH(1,1)-Normal**, pass all four Kupiec and Christoffersen tests. The result that Normal-distribution GARCH models pass backtesting while Student- $t$  models fail at the 95% level highlights an important distinction between in-sample fit and out-of-sample risk calibration. The Student- $t$  models achieve substantially better AIC/BIC scores and are theoretically better suited to fat-tailed returns, but their quantile estimates at the 5% left tail are sufficiently extreme that actual violations fall significantly below the expected rate. For regulatory purposes at the 95% level, the simpler Normal models are better calibrated. At the 99% level, however, the Student- $t$  models perform well, and the EGARCH(1,1)-Student- $t$  specifically is the only model that comfortably passes at the level most directly relevant to Basel III capital requirements. Practitioners should therefore consider the confidence level for which the model is being calibrated when choosing between Gaussian and fat-tailed specifications.

Figure 15 synthesizes the three key evaluation dimensions of the study into a single three-panel summary chart, providing a consolidated view of the trade-offs across all models.

### 4. Stress-Period Robustness Analysis

Table 22 disaggregates the full-sample backtesting results into two qualitatively distinct subperiods: the 2022 bear market ( $T = 251$ ) and the subsequent 2023–24 recovery ( $T = 501$ ). This decomposition reveals that full-sample pass/fail outcomes mask substantial heterogeneity across market regimes.

GARCH(1,1)-Normal and EGARCH(1,1)-Normal, the only models achieving a full-sample pass, both fail the Kupiec test at the 95% level during 2022 (violation rates of 9.16% and 10.36% against a nominal 5.00%). Their aggregate pass is therefore attributable to error cancellation between the bear market and the calmer recovery, rather than to genuine robustness. These models are correctly calibrated *on average*, but not during the stress episodes when VaR accuracy is most consequential.

The Student- $t$  specifications exhibit the opposite pattern: conservative during stress but over-conservative in calm periods. GARCH(1,1)-Student- $t$  and GJR-GARCH(1,1)-Student- $t$  pass the Kupiec test during 2022 at both confidence levels, yet fail at the 95% level during the recovery (violation rates of 2.59% and 2.40%), as their heavy-tailed quantiles prove unnecessarily wide in a low-volatility environment. EGARCH(1,1)-Student- $t$  presents the most favourable profile: it passes at both confidence levels in 2022 (95%:  $p = 0.873$ ; 99%:  $p = 0.276$ ) and at the 99% level across all periods, supporting its recommended use for Basel III capital adequacy assessment.

HAR-RV fails the Kupiec test in both subperiods at both confidence levels ( $p < 0.001$  in all cases), with violation rates of 14.74% and 9.18% at the 95% level during the bear market and recovery respectively. The persistence of failure across regimes confirms that HAR-RV's miscalibration is structural: the combination of an absolute-return proxy

and Normal quantiles produces VaR boundaries that are systematically too shallow regardless of prevailing volatility conditions.

The subsample analysis yields a three-way typology of calibration behaviour. Normal GARCH models exhibit *regime-conditional* accuracy, passing on average but failing under stress. Student-*t* models exhibit *conservative asymmetry*, reliable during stress but over-conservative in calm periods, with EGARCH(1,1)-Student-*t* offering the most favourable regulatory profile. HAR-RV exhibits *uniform structural bias*, failing across all regimes and confidence levels. These findings reinforce the view that model selection should be explicitly conditioned on the intended application and target market regime.

## VI. Conclusion

### A. Summary of Findings

This study examined seven competing volatility models, comprising six GARCH-family specifications and the HAR-RV benchmark, on fifteen years of daily S&P 500 returns. All models were evaluated within a unified expanding-window framework that covers two qualitatively distinct stress regimes: the COVID-19 market crash of 2020 and the Federal Reserve rate-hiking cycle of 2022. The analysis progressed through in-sample estimation and model selection, out-of-sample volatility forecasting, VaR and ES computation, and formal backtesting. Five main findings are reported below.

*First*, the standard preconditions for GARCH modelling are clearly satisfied by the data. The log-return series is confirmed to be  $I(0)$  stationary by both the ADF test (stat =  $-13.34$ ,  $p < 0.001$ ) and the KPSS test (stat =  $0.027$ ,  $p \geq 0.10$ ). The ARCH-LM test applied to the ARIMA(4, 0, 0) residuals yields a  $\chi^2$  statistic of  $982.20$  ( $p < 0.001$ ), confirming strong and persistent conditional heteroskedasticity that warrants the use of GARCH-type variance models.

*Second*, EGARCH(1,1)-Student-*t* is selected as the best in-sample model under both AIC (7,191) and BIC (7,227). The estimated parameters have clear economic interpretations. The GARCH coefficient  $\hat{\beta} = 0.959$  indicates very high volatility persistence, implying that shocks to conditional variance take approximately three to four months to decay to the long-run level. The leverage parameter  $\hat{\gamma} = -0.211$  ( $p < 0.001$ ) confirms that negative return shocks raise conditional variance by a greater amount than positive shocks of equal size, consistent with the well-known leverage effect in equity markets. The estimated degrees-of-freedom parameter  $\hat{\nu} = 5.49$  ( $p < 0.001$ ) reflects economically significant fat tails that the Gaussian distribution cannot adequately represent. Across all three variance specifications, Student-*t* models uniformly outperform their Gaussian counterparts, with AIC improvements ranging from 210 to 230 points.

*Third*, HAR-RV achieves the best out-of-sample point forecast accuracy among all seven models under both MSE (0.504) and MAE (0.542). The best-performing GARCH model under MAE is EGARCH(1,1)-Normal (0.586), which represents an 8.1% deterioration relative to HAR-RV. This result is noteworthy given that HAR-RV uses only absolute daily returns as its realized volatility proxy rather than high-frequency intraday data. The competitive forecasting performance of HAR-RV reflects its ability to aggregate volatility signals at daily, weekly, and monthly horizons, thereby capturing the heterogeneous decision horizons of different market participants without requiring any distributional assumption. However, under the QLIKE loss function, which penalizes under-prediction of volatility more heavily than over-prediction, the ranking reverses. EGARCH(1,1)-Normal records the lowest QLIKE (0.988), while HAR-RV produces the highest (1.306) among all models. This divergence suggests that HAR-RV tends to underestimate volatility during high-risk periods, precisely when accurate calibration carries the greatest practical importance.

*Fourth*, the backtesting results expose a fundamental limitation of the HAR-RV model for risk management applications. The model fails the Kupiec Proportion of Failures test at both the 95% and 99% confidence levels ( $p < 0.001$ ), recording empirical violation rates of 10.99% and 4.77% against the corresponding nominal rates of 5.00% and 1.00%. This outcome is attributable to a structural bias: the use of average absolute returns as the volatility proxy, combined with Normal distribution quantiles, produces VaR boundaries that are systematically too shallow to cover the actual left tail of the return distribution, particularly during the prolonged bear market of 2022. In contrast, GARCH(1,1)-Normal and EGARCH(1,1)-Normal pass all four Kupiec and Christoffersen tests, demonstrating both correct unconditional coverage and temporal independence of violations across both confidence levels. Notably, all seven models pass the Christoffersen independence test, indicating that no specification generates clusters of consecutive VaR violations, a key requirement for regulatory compliance under Basel III.

*Fifth*, the distributional assumption interacts with the target confidence level in a practically important way. Student-*t* specifications achieve superior in-sample fit and generate more conservative risk estimates. For example,



EGARCH(1,1)-Student- $t$  produces an average VaR at the 99% level of  $-3.52\%$ , compared with  $-1.85\%$  for HAR-RV. At the 99% confidence level, Student- $t$  models pass the Kupiec test, making them appropriate for Basel III capital adequacy assessment. At the 95% level, however, the same models fail due to over-conservatism, with empirical violation rates of approximately 3.1% falling well below the nominal 5.0%. Gaussian models, despite their inferior in-sample fit, are better calibrated at the 95% level and achieve a 100% pass rate across all backtesting criteria.

*Sixth*, the subsample analysis qualifies the full-sample backtesting conclusions in an important respect. GARCH(1,1)-Normal and EGARCH(1,1)-Normal, the only models to pass all full-sample backtests, both fail the Kupiec test during the 2022 bear market with violation rates exceeding 9% at the 95% level. Their aggregate pass reflects partial cancellation of excess stress-period violations against near-correct recovery-period coverage, rather than genuine robustness across market regimes. Student- $t$  specifications exhibit the complementary pattern, passing during stress periods while failing during calm periods, with EGARCH(1,1)-Student- $t$  offering the most balanced profile for regulatory use. HAR-RV fails uniformly in both subperiods, confirming that its calibration bias is structural in nature rather than episodic.

## B. Practical Implications

The results of this study offer several actionable insights for risk practitioners. For regulatory capital reporting under Basel III, where Expected Shortfall at the 99% level is the mandated risk measure, asymmetric GARCH models with Student- $t$  innovations represent the most appropriate choice. These models capture the leverage effect, assign sufficient probability mass to extreme losses through their fat-tailed quantiles, and pass the regulatory backtesting requirements at the 99% confidence level. EGARCH(1,1)-Student- $t$ , as the best-fitting model in-sample, is the natural recommendation for this purpose.

For internal risk monitoring applications where the 95% confidence level is used, Gaussian GARCH models such as GARCH(1,1)-Normal or EGARCH(1,1)-Normal offer better-calibrated VaR estimates and avoid the unnecessary capital reserves that would arise from the over-conservatism of Student- $t$  specifications at this level.

The HAR-RV model, in its current form, should not be used directly for regulatory VaR reporting when only daily closing prices are available. Its violation rates at both confidence levels are more than twice the nominal values, which would render a risk model non-compliant with the backtesting standards required under Basel III. That said, HAR-RV remains a valuable tool for monitoring the general level and direction of conditional volatility. Its strong point-forecast accuracy under symmetric loss functions makes it a natural component of hybrid modelling approaches that combine HAR-RV volatility predictions with a parametric distributional assumption for tail risk quantification.

## C. Limitations and Future Research

Three limitations of the present study should be acknowledged. First, realized volatility is proxied throughout by the absolute daily log return  $|r_t|$ , following the approach of Ding et al. [16]. Although this proxy preserves the multi-horizon structure of the HAR-RV model and is appropriate when intraday data are unavailable, it is a noisy one-observation estimator that discards all within-day price variation. The use of high-frequency realized variance, as established by Andersen and Bollerslev [15], would eliminate this source of measurement error and allow a more precise comparison between the two modelling paradigms.

Second, the empirical analysis is conducted on a single equity index. The relative performance of GARCH and HAR-RV models may vary across different asset classes, including commodity markets, foreign exchange rates, and fixed income securities, and across market environments not represented in the 2010 to 2024 sample window.

Third, all models in this study are estimated within a univariate framework. Practical portfolio risk management requires multivariate extensions. Examples include the Dynamic Conditional Correlation GARCH model for capturing time-varying co-movements across assets, and the GARCH-MIDAS framework for incorporating low-frequency macroeconomic variables into the variance equation. These extensions are left for future research.

Future work may also investigate machine learning hybrid approaches that retain the point-forecast accuracy of HAR-RV while correcting its tendency to underestimate tail risk. Promising directions include LSTM-GARCH architectures and conformal prediction intervals. Additionally, the use of jump-robust realized measures, such as the bipower variation estimator of Barndorff-Nielsen and Shephard [19], would allow the decomposition of realized volatility into its continuous and jump components, potentially improving both forecast accuracy and VaR calibration during periods of acute market stress of the kind documented in this study.

## References

- [1] Mandelbrot, B., “The Variation of Certain Speculative Prices,” *Journal of Business*, Vol. 36, No. 4, 1963, pp. 394–419. <https://doi.org/10.1086/294632>.
- [2] Engle, R. F., “Autoregressive Conditional Heteroscedasticity with Estimates of the Variance of United Kingdom Inflation,” *Econometrica*, Vol. 50, No. 4, 1982, pp. 987–1007. <https://doi.org/10.2307/1912773>.
- [3] Bollerslev, T., “Generalized Autoregressive Conditional Heteroskedasticity,” *Journal of Econometrics*, Vol. 31, No. 3, 1986, pp. 307–327. [https://doi.org/10.1016/0304-4076\(86\)90063-1](https://doi.org/10.1016/0304-4076(86)90063-1).
- [4] Glosten, L. R., Jagannathan, R., and Runkle, D. E., “On the Relation between the Expected Value and the Volatility of the Nominal Excess Return on Stocks,” *Journal of Finance*, Vol. 48, No. 5, 1993, pp. 1779–1801. <https://doi.org/10.1111/j.1540-6261.1993.tb05128.x>.
- [5] Nelson, D. B., “Conditional Heteroskedasticity in Asset Returns: A New Approach,” *Econometrica*, Vol. 59, No. 2, 1991, pp. 347–370. <https://doi.org/10.2307/2938260>.
- [6] Andersen, T. G., and Bollerslev, T., “Answering the Skeptics: Yes, Standard Volatility Models Do Provide Accurate Forecasts,” *International Economic Review*, Vol. 39, No. 4, 1998, pp. 885–905. <https://doi.org/10.2307/2527343>.
- [7] Corsi, F., “A Simple Approximate Long-Memory Model of Realized Volatility,” *Journal of Financial Econometrics*, Vol. 7, No. 2, 2009, pp. 174–196. <https://doi.org/10.1093/jjfinec/nbp001>.
- [8] Müller, U. A., Dacorogna, M. M., Davé, R. D., Olsen, R. B., Pictet, O. V., and von Weizsäcker, J. E., “Volatilities of Different Time Resolutions: Analysing the Dynamics of Market Components,” *Journal of Empirical Finance*, Vol. 4, No. 2–3, 1997, pp. 213–239. [https://doi.org/10.1016/S0927-5398\(97\)00007-8](https://doi.org/10.1016/S0927-5398(97)00007-8).
- [9] Artzner, P., Delbaen, F., Eber, J.-M., and Heath, D., “Coherent Measures of Risk,” *Mathematical Finance*, Vol. 9, No. 3, 1999, pp. 203–228. <https://doi.org/10.1111/1467-9965.00068>.
- [10] Basel Committee on Banking Supervision, “Minimum Capital Requirements for Market Risk,” Tech. rep., Bank for International Settlements, Basel, Switzerland, 2016. URL <https://www.bis.org/bcbs/publ/d352.htm>.
- [11] Kupiec, P. H., “Techniques for Verifying the Accuracy of Risk Measurement Models,” *Journal of Derivatives*, Vol. 3, No. 2, 1995, pp. 73–84. <https://doi.org/10.3905/jod.1995.407942>.
- [12] Christoffersen, P. F., “Evaluating Interval Forecasts,” *International Economic Review*, Vol. 39, No. 4, 1998, pp. 841–862. <https://doi.org/10.2307/2527341>.
- [13] Glosten, L. R., Jagannathan, R., and Runkle, D. E., “On the Relation between the Expected Value and the Volatility of the Nominal Excess Return on Stocks,” *The Journal of Finance*, Vol. 48, No. 5, 1993, pp. 1779–1801. <https://doi.org/10.1111/j.1540-6261.1993.tb05128.x>.
- [14] Bollerslev, T., “A Conditionally Heteroskedastic Time Series Model for Speculative Prices and Rates of Return,” *Review of Economics and Statistics*, Vol. 69, No. 3, 1987, pp. 542–547. <https://doi.org/10.2307/1925546>.
- [15] Andersen, T. G., and Bollerslev, T., “Answering the Skeptics: Yes, Standard Volatility Models Do Provide Accurate Forecasts,” *International Economic Review*, Vol. 39, No. 4, 1998, pp. 885–905. <https://doi.org/10.2307/2527343>.
- [16] Ding, Z., Granger, C. W. J., and Engle, R. F., “A long memory property of stock market returns and a new model,” *Journal of Empirical Finance*, Vol. 1, No. 1, 1993, pp. 83–106. [https://doi.org/10.1016/0927-5398\(93\)90006-D](https://doi.org/10.1016/0927-5398(93)90006-D).
- [17] Dickey, D. A., and Fuller, W. A., “Distribution of the Estimators for Autoregressive Time Series with a Unit Root,” *Journal of the American Statistical Association*, Vol. 74, No. 366, 1979, pp. 427–431. <https://doi.org/10.1080/01621459.1979.10482531>.
- [18] Kwiatkowski, D., Phillips, P. C. B., Schmidt, P., and Shin, Y., “Testing the Null Hypothesis of Stationarity against the Alternative of a Unit Root: How Sure Are We That Economic Time Series Have a Unit Root?” *Journal of Econometrics*, Vol. 54, No. 1–3, 1992, pp. 159–178. [https://doi.org/10.1016/0304-4076\(92\)90104-Y](https://doi.org/10.1016/0304-4076(92)90104-Y).
- [19] Barndorff-Nielsen, O. E., and Shephard, N., “Power and Bipower Variation with Stochastic Volatility and Jumps,” *Journal of Financial Econometrics*, Vol. 2, No. 1, 2004, pp. 1–37. <https://doi.org/10.1093/jjfinec/nbh001>.
- [20] Diebold, F. X., and Mariano, R. S., “Comparing Predictive Accuracy,” *Journal of Business & Economic Statistics*, Vol. 13, No. 3, 1995, pp. 253–263. <https://doi.org/10.1080/07350015.1995.10524599>.
- [21] McNeil, A. J., and Frey, R., “Estimation of Tail-Related Risk Measures for Heteroscedastic Financial Time Series: An Extreme Value Approach,” *Journal of Empirical Finance*, Vol. 7, 2000, pp. 271–300. [https://doi.org/10.1016/S0927-5398\(00\)00012-8](https://doi.org/10.1016/S0927-5398(00)00012-8).

## Appendix A Full Parameter and Results Tables

This appendix consolidates the complete numerical output for all models estimated in this study. Table 12 presents the full GARCH-family in-sample estimates. Table 13 details the ARIMA(4, 0, 0) mean-equation results. Table 14 provides the complete out-of-sample forecast accuracy ranking (MSE, MAE, and QLIKE) for all seven models. Table 15 gives the full VaR and Expected Shortfall summary. Table 16 is the complete backtesting report. Table 17 provides the pass/fail summary. Table 18 reproduces the HAR-RV OLS full output. Table 19 reports the Diebold-Mariano forecast accuracy tests. Table 20 reports the Expected Shortfall backtesting results. Table 21 presents GARCH standardised residual diagnostics. Table 22 presents subsample backtesting results across distinct market regimes (2022 bear market and 2023–24 recovery). Table 23 presents the ARIMA mean-equation specification comparison and robustness check. Table 24 reports the restricted GJR-GARCH likelihood ratio test for the  $\alpha = 0$  boundary solution.

**Table 12 GARCH-Family In-Sample Estimation Results (Training set, January 2010 – December 2021,  $T = 3,017$ ). Parameters:  $\omega$  = variance intercept;  $\alpha$  = ARCH coefficient;  $\beta$  = GARCH coefficient;  $\gamma$  = asymmetry (leverage) parameter;  $\nu$  = Student- $t$  degrees of freedom. For EGARCH, Persistence =  $|\beta|$  (values  $> 1$  are permissible); for GARCH/GJR, Persistence =  $\alpha + \beta$  (GJR:  $\alpha + \beta + 0.5\gamma$ ). \*\*\*  $p < 0.001$ ; \*\*  $p < 0.01$ ; \*  $p < 0.05$ ; n.s. = not significant. ★ = best model by AIC and BIC.**

| Model                | Dist.                         | $\omega$                      | $\alpha$               | $\beta$          | $\gamma$          | $\nu$            | AIC             | BIC             | Persist.                  | Log- $L$         |
|----------------------|-------------------------------|-------------------------------|------------------------|------------------|-------------------|------------------|-----------------|-----------------|---------------------------|------------------|
| GARCH(1,1)           | Normal                        | 0.0421***                     | 0.1887***              | 0.7745***        | —                 | —                | 7,531.25        | 7,555.30        | 0.9632                    | −3,761.62        |
| GARCH(1,1)           | Student- $t$                  | 0.0293***                     | 0.1924***              | 0.7973***        | —                 | 4.9991***        | 7,320.70        | 7,350.76        | 0.9897                    | −3,655.35        |
| GJR-GARCH(1,1)       | Normal                        | 0.0429***                     | 0.0439 <sup>n.s.</sup> | 0.7854***        | 0.2590***         | —                | 7,432.96        | 7,463.02        | 0.9587                    | −3,711.48        |
| GJR-GARCH(1,1)       | Student- $t$                  | 0.0330***                     | 0.0000 <sup>†</sup>    | 0.8066***        | 0.3353***         | 5.3233***        | 7,202.80        | 7,238.87        | 0.9742                    | −3,595.40        |
| EGARCH(1,1)          | Normal                        | −0.0091 <sup>n.s.</sup>       | 0.2484***              | 0.9426***        | −0.1743***        | —                | 7,408.99        | 7,439.05        | 0.9426 <sup>‡</sup>       | −3,699.50        |
| <b>EGARCH(1,1) ★</b> | <b>Student-<math>t</math></b> | <b>−0.0118<sup>n.s.</sup></b> | <b>0.2047***</b>       | <b>0.9594***</b> | <b>−0.2107***</b> | <b>5.4906***</b> | <b>7,191.03</b> | <b>7,227.10</b> | <b>0.9594<sup>‡</sup></b> | <b>−3,589.51</b> |

<sup>†</sup> GJR-GARCH(1,1)-Student- $t$ :  $\alpha$  estimate is effectively zero (numerical boundary); leverage effect fully captured by  $\gamma = 0.3353$ \*\*\*.

<sup>‡</sup> For EGARCH, Persistence =  $|\beta|$  from the log-variance equation; covariance stationarity is governed by a different condition so the displayed value differs from the GARCH/GJR definition.

Significance stars based on two-sided  $z$ -tests from Maximum Likelihood estimation using the arch library.

**Table 13 ARIMA(4, 0, 0) Mean-Equation Estimates (Training set,  $T = 3,017$ , January 2010 – December 2021). Selected by stepwise AIC minimisation (auto\_arima, pmdarima). Dependent variable:  $r_t = \ln(P_t/P_{t-1}) \times 100$ . \*\*\*  $p < 0.001$ ; \*  $p < 0.05$ .**

| Parameter                              | Coefficient | Std. Error | $t$ -stat | $p$ -value | Sig. |
|----------------------------------------|-------------|------------|-----------|------------|------|
| $\hat{\mu}$ (Intercept)                | 0.0540      | 0.0219     | 2.4692    | 0.0135     | *    |
| $\hat{\phi}_1$ AR(1)                   | −0.1372     | 0.0084     | −16.2825  | < 0.001    | ***  |
| $\hat{\phi}_2$ AR(2)                   | 0.0830      | 0.0075     | 11.0424   | < 0.001    | ***  |
| $\hat{\phi}_3$ AR(3)                   | −0.0176     | 0.0086     | −2.0574   | 0.0396     | *    |
| $\hat{\phi}_4$ AR(4)                   | −0.0606     | 0.0089     | −6.8362   | < 0.001    | ***  |
| $\hat{\sigma}^2$ (Innovation variance) | 1.1411      | 0.0122     | 93.4808   | < 0.001    | ***  |
| AIC                                    | 8,972.27    |            | —         |            |      |
| BIC                                    | 9,008.34    |            | —         |            |      |

All parameters estimated by Maximum Likelihood via auto\_arima (pmdarima library). Standard errors are heteroskedasticity-robust. The positive intercept ( $\hat{\mu} = 0.054$ ) reflects the positive unconditional mean return of the S&P 500 over the sample period.

$\hat{\sigma}^2$  is the estimated unconditional innovation variance reported by the ARIMA fit; it is superseded by the time-varying conditional variance from the subsequent GARCH stage.

**Table 14 Out-of-Sample Forecast Accuracy: MSE, MAE, and QLIKE (Test set,  $T = 755$  observations, December 2021 – December 2024). Realized volatility proxy:  $RV_t = |r_t|$ . Lower values indicate better performance for all three metrics. Bold entries indicate the best value for each metric.**

| Model          | Dist.        | MSE           | MAE           | QLIKE         | MAE Rank |
|----------------|--------------|---------------|---------------|---------------|----------|
| <b>HAR-RV</b>  | —            | <b>0.5041</b> | <b>0.5421</b> | 1.3059        | <b>1</b> |
| EGARCH(1,1)    | Normal       | 0.5503        | 0.5857        | <b>0.9881</b> | 2        |
| EGARCH(1,1) ★  | Student- $t$ | 0.5679        | 0.5887        | 0.9912        | 3        |
| GJR-GARCH(1,1) | Normal       | 0.5734        | 0.5910        | 1.0053        | 4        |
| GJR-GARCH(1,1) | Student- $t$ | 0.5994        | 0.5965        | 1.0057        | 5        |
| GARCH(1,1)     | Normal       | 0.5743        | 0.6054        | 1.0283        | 6        |
| GARCH(1,1)     | Student- $t$ | 0.5936        | 0.6143        | 1.0244        | 7        |

HAR-RV achieves the lowest MSE and MAE but the highest QLIKE (1.306), indicating systematic underestimation of tail volatility. Best QLIKE: EGARCH(1,1)-Normal (0.988). See Table 19 for Diebold-Mariano tests of statistical significance. MSE and MAE use  $RV_t = |r_t|$  as proxy; QLIKE penalises under-prediction of variance more heavily.

**Table 15 Average Daily VaR and Expected Shortfall on the Test Set (December 2021 – December 2024,  $T = 755$ ). Negative values represent expected losses. Expected violation rates: 5.00% at 95% CL; 1.00% at 99% CL. ★ = best in-sample model.**

| Model          | Dist.        | Avg VaR 95% | Avg VaR 99% | Avg ES 99% | Violations 95% | Violations 99% |
|----------------|--------------|-------------|-------------|------------|----------------|----------------|
| GARCH(1,1)     | Normal       | −1.7046%    | −2.4109%    | −2.7620%   | 49 (6.49%)     | 13 (1.72%)     |
| GARCH(1,1)     | Student- $t$ | −2.1514%    | −3.5927%    | −4.7538%   | 23 (3.05%)     | 3 (0.40%)      |
| GJR-GARCH(1,1) | Normal       | −1.6848%    | −2.3828%    | −2.7299%   | 55 (7.28%)     | 17 (2.25%)     |
| GJR-GARCH(1,1) | Student- $t$ | −2.1061%    | −3.5170%    | −4.6536%   | 24 (3.18%)     | 3 (0.40%)      |
| EGARCH(1,1)    | Normal       | −1.6868%    | −2.3857%    | −2.7332%   | 50 (6.62%)     | 13 (1.72%)     |
| EGARCH(1,1) ★  | Student- $t$ | −2.1097%    | −3.5230%    | −4.6615%   | 23 (3.05%)     | 4 (0.53%)      |
| HAR-RV         | Normal       | −1.3063%    | −1.8476%    | −2.1167%   | 83 (10.99%)    | 36 (4.77%)     |

VaR and ES computed parametrically using expanding-window conditional volatility forecasts. Student- $t$  specifications are systematically more conservative owing to heavier quantiles at  $\hat{\nu} \approx 5.0$ – $5.5$ .

HAR-RV applies Normal quantiles to  $|r_t|$  as the volatility proxy, producing the least conservative risk estimates of all seven models.

ES backtesting results are reported in Table 20.

*End of Appendix A*

**Table 16 Backtesting Results: Kupiec Proportion of Failures (POF) Test and Christoffersen Conditional Coverage (CC) Test (Test set,  $T = 755$ ). Expected violation rates: 5.00 % at 95 % CL; 1.00 % at 99 % CL. Pass criterion:  $p$ -value  $> 0.05$ . ★ = best in-sample model. Table 10 in the paper body.**

| Model                       | CL | Exp. N | Exp. % | Act. % | $N_v$ | Kupiec $p$ | Kupiec | Christoff. $p$ | Christoff. |
|-----------------------------|----|--------|--------|--------|-------|------------|--------|----------------|------------|
| GARCH(1,1)-Normal           | 95 | 37.75  | 5.00   | 6.49   | 49    | 0.072      | Pass   | 0.450          | Pass       |
|                             | 99 | 7.55   | 1.00   | 1.72   | 13    | 0.071      | Pass   | 0.499          | Pass       |
| GARCH(1,1)-Student- $t$     | 95 | 37.75  | 5.00   | 3.05   | 23    | 0.008      | Fail   | 0.229          | Pass       |
|                             | 99 | 7.55   | 1.00   | 0.40   | 3     | 0.058      | Pass   | 0.877          | Pass       |
| GJR-GARCH(1,1)              | 95 | 37.75  | 5.00   | 7.28   | 55    | 0.007      | Fail   | 0.571          | Pass       |
|                             | 99 | 7.55   | 1.00   | 2.25   | 17    | 0.003      | Fail   | 0.376          | Pass       |
| GJR-GARCH(1,1)-Student- $t$ | 95 | 37.75  | 5.00   | 3.18   | 24    | 0.014      | Fail   | 0.789          | Pass       |
|                             | 99 | 7.55   | 1.00   | 0.40   | 3     | 0.058      | Pass   | 0.877          | Pass       |
| EGARCH(1,1)                 | 95 | 37.75  | 5.00   | 6.62   | 50    | 0.051      | Pass   | 0.851          | Pass       |
|                             | 99 | 7.55   | 1.00   | 1.72   | 13    | 0.071      | Pass   | 0.499          | Pass       |
| EGARCH(1,1)                 | 95 | 37.75  | 5.00   | 3.05   | 23    | 0.008      | Fail   | 0.229          | Pass       |
|                             | 99 | 7.55   | 1.00   | 0.53   | 4     | 0.154      | Pass   | 0.836          | Pass       |
| HAR-RV                      | 95 | 37.75  | 5.00   | 10.99  | 83    | 0.000      | Fail   | 0.959          | Pass       |
|                             | 99 | 7.55   | 1.00   | 4.77   | 36    | 0.000      | Fail   | 0.535          | Pass       |

Green = pass ( $p > 0.05$ ); Red = fail ( $p \leq 0.05$ ).

All seven models pass the Christoffersen independence test ( $p \geq 0.229$ ), confirming no temporally clustered violations.

HAR-RV fails Kupiec at both levels ( $p = 0.000$ ) due to systematic level bias.

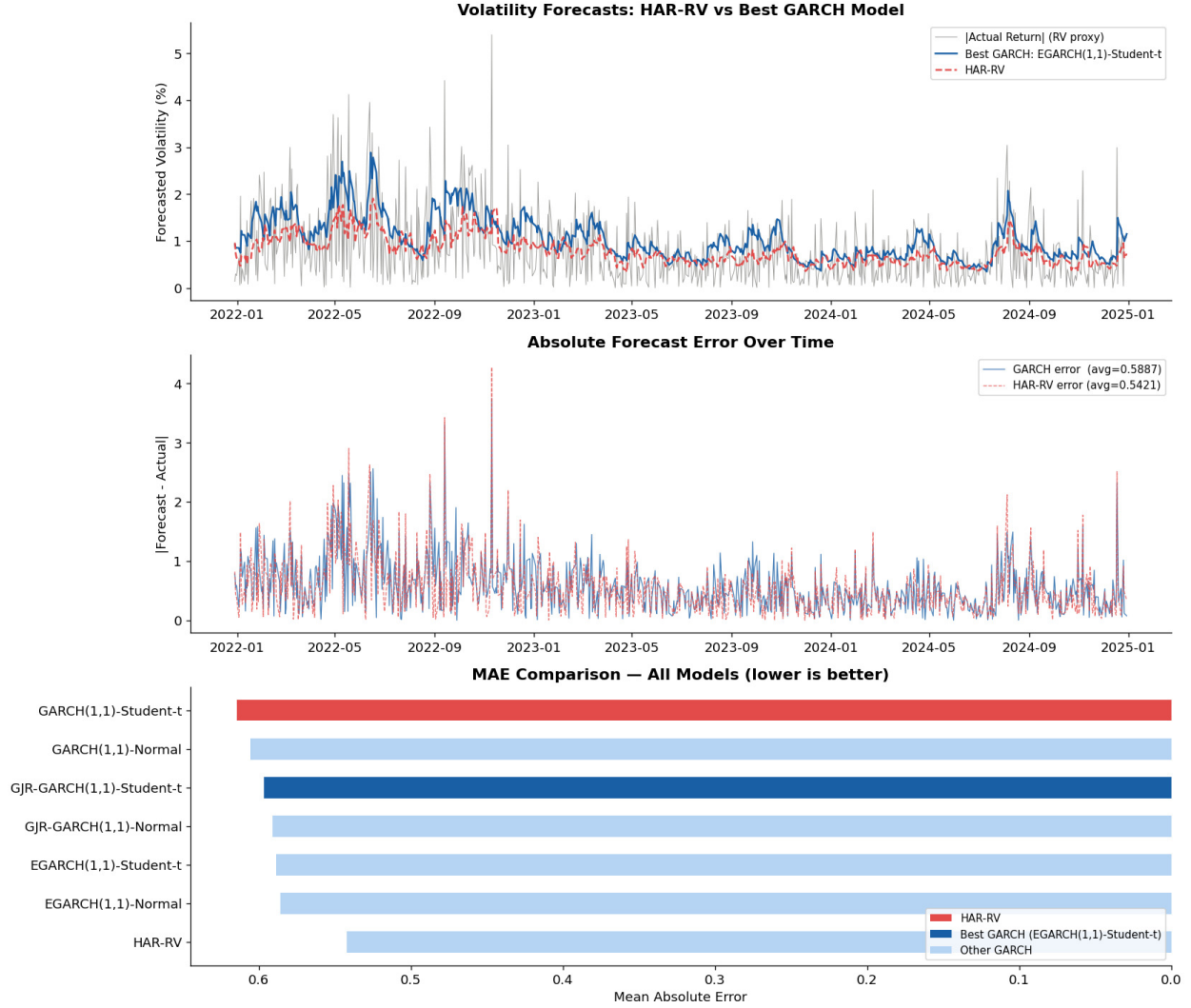
**Table 17 Backtesting Pass/Fail Summary. A model achieves an overall pass (✓) if it satisfies both the Kupiec and Christoffersen tests at both confidence levels. Source: Table 11 in the paper body.**

| Model                       | Kupiec 95% | Kupiec 99% | Christoff. (all) | Overall |
|-----------------------------|------------|------------|------------------|---------|
| GARCH(1,1)-Normal           | Pass       | Pass       | Pass             | ✓       |
| GARCH(1,1)-Student- $t$     | Fail       | Pass       | Pass             | Partial |
| GJR-GARCH(1,1)-Normal       | Fail       | Fail       | Pass             | ×       |
| GJR-GARCH(1,1)-Student- $t$ | Fail       | Pass       | Pass             | Partial |
| EGARCH(1,1)-Normal          | Pass       | Pass       | Pass             | ✓       |
| EGARCH(1,1)-Student- $t$ ★  | Fail       | Pass       | Pass             | Partial |
| HAR-RV                      | Fail       | Fail       | Pass             | ×       |

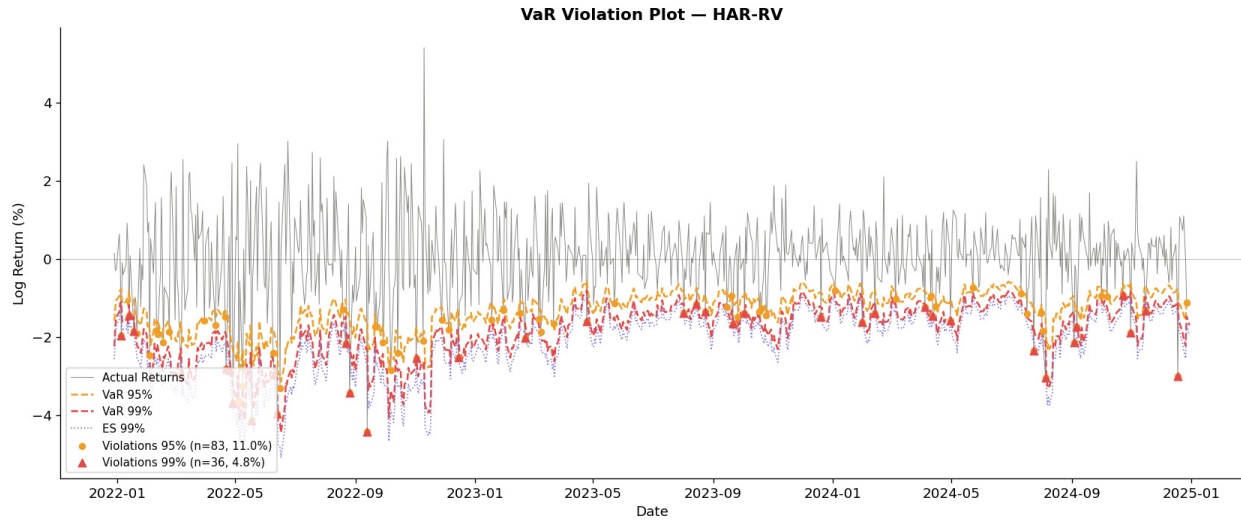
Green rows achieve a full pass (✓).

“Partial” = passes Kupiec at 99 % but fails at 95 % (over-conservatism from Student- $t$  heavy tails at the 5 % quantile).

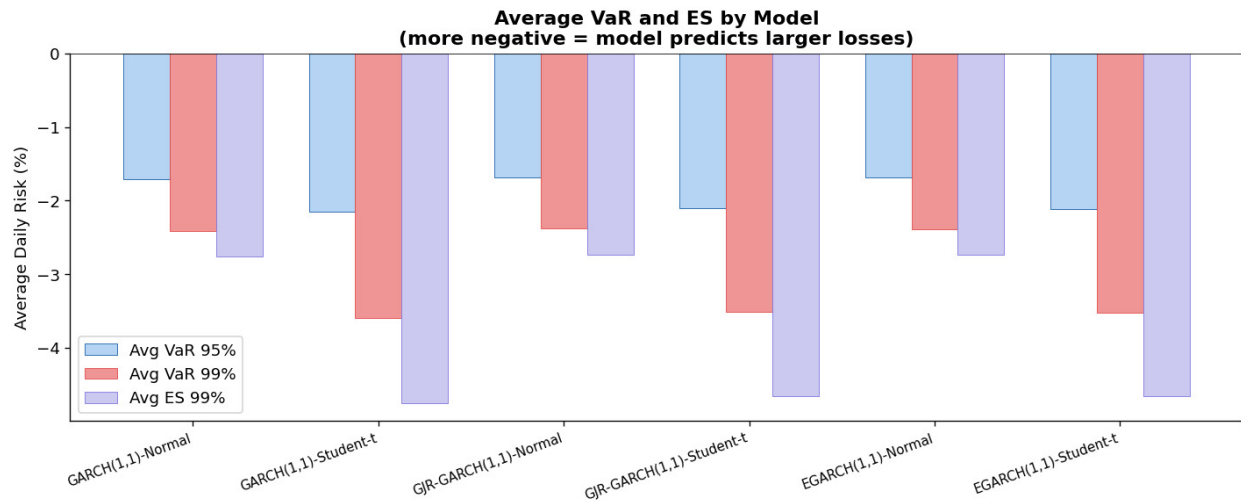
For regulatory purposes (Basel III, 99 % ES), EGARCH(1,1)-Student- $t$  is the preferred model despite its 95 % failure.



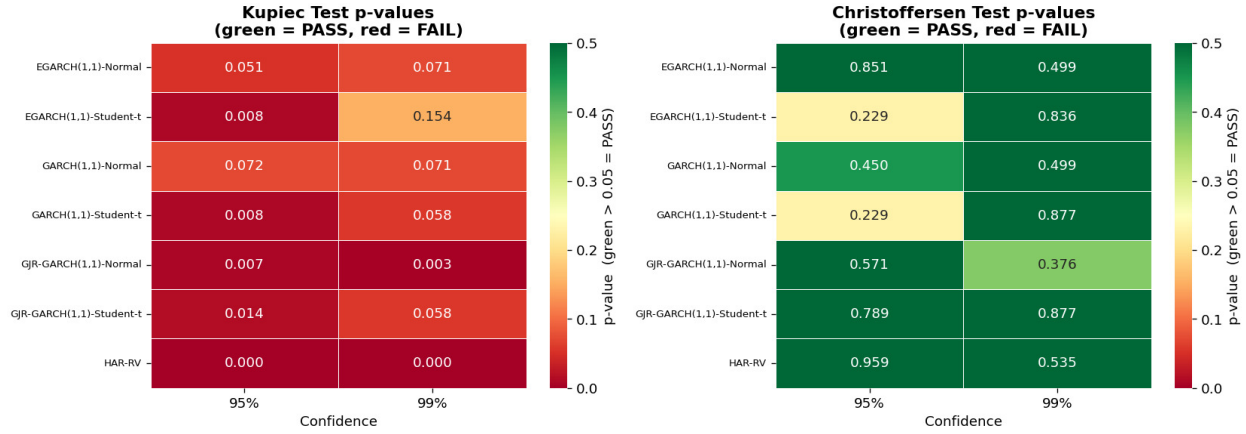
**Fig. 11 HAR-RV vs. EGARCH(1,1)-Student- $t$ : Three-Panel Forecast Comparison (Test Set, December 2021 – December 2024).** *Upper panel:* One-step-ahead volatility forecasts from EGARCH(1,1)-Student- $t$  (blue solid) and HAR-RV (red dashed) against the actual realized volatility proxy  $|r_t|$  (grey). The GARCH model produces higher and more reactive forecasts, especially during stress episodes; the HAR-RV forecast is smoother and systematically lower. *Middle panel:* Absolute forecast error over time for both models. HAR-RV (red dashed, avg MAE = 0.542) achieves smaller errors on most days; GARCH errors (blue solid, avg MAE = 0.589) spike more sharply during volatility peaks but are comparable in calm periods. *Lower panel:* MAE bar chart for all seven models. HAR-RV achieves the lowest MAE; GARCH(1,1)-Student- $t$  the highest.



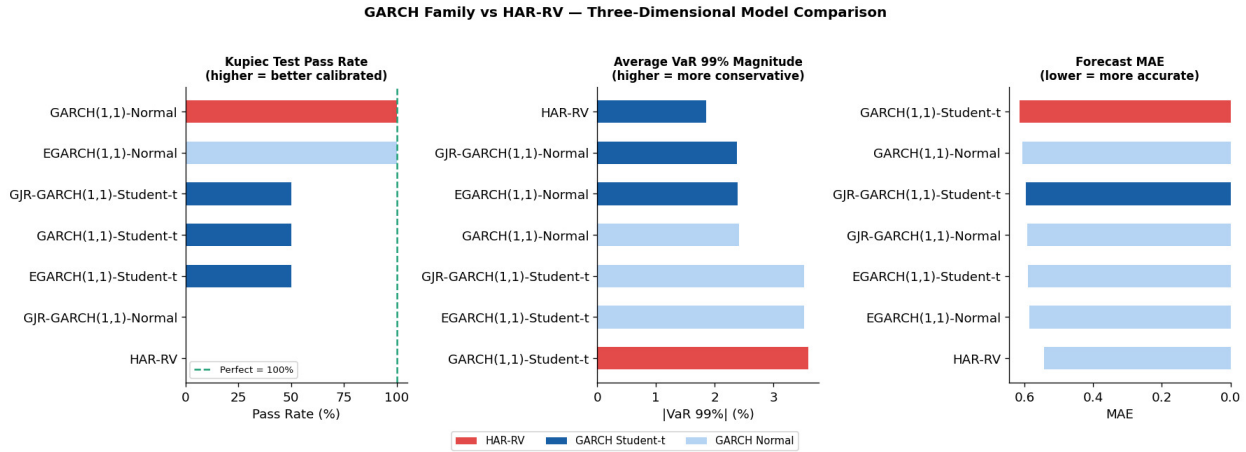
**Fig. 12 VaR Violation Plot – HAR-RV Model (Test Set, December 2021 – December 2024).** Grey line: actual daily log returns. Orange dashed: VaR 95% boundary. Red dashed: VaR 99% boundary. Blue dotted: ES 99%. Orange circles: 95% violations ( $n = 83$ , rate = 11.0%, expected 5.0%). Red triangles: 99% violations ( $n = 36$ , rate = 4.8%, expected 1.0%). The dense cluster of violation markers throughout the test period, particularly during 2022, confirms that the HAR-RV's VaR boundaries are systematically too shallow to capture the true loss distribution.



**Fig. 13 Average Daily VaR and ES by Model (Test Set, December 2021 – December 2024).** Grouped bars show average VaR 95% (blue), VaR 99% (red), and ES 99% (purple) for each model. More negative values indicate more conservative risk estimates. Within each model class, the Student- $t$  specification (right bar of each pair) is substantially more negative than the Normal specification (left bar), confirming the systematic impact of the fat-tail assumption on tail-risk quantification. HAR-RV is excluded from this figure as it does not appear alongside the GARCH models in this output; its average estimates (VaR 99% = -1.85%, ES 99% = -2.12%) are well above the range shown here.



**Fig. 14** Backtesting  $p$ -value Heatmaps: Kupiec POF Test (left) and Christoffersen Conditional Coverage Test (right). Rows: models. Columns: confidence levels (95%, 99%). Color scale: green =  $p > 0.05$  (Pass); red =  $p < 0.05$  (Fail). *Kupiec heatmap*: EGARCH(1,1)-Normal and GARCH(1,1)-Normal are the only models with all green cells; HAR-RV has the darkest red ( $p = 0.000$  at both levels), confirming severe miscalibration. *Christoffersen heatmap*: All models pass across all confidence levels ( $p \geq 0.229$ ), indicating that no model generates temporally clustered violations.



**Fig. 15** Three-Dimensional Model Comparison: Kupiec Pass Rate, Average VaR 99% Magnitude, and Forecast MAE (Test Set, December 2021 – December 2024). *Left panel*: Kupiec test pass rate (% of tests passed across both confidence levels; higher = better calibrated). GARCH(1,1)-Normal and EGARCH(1,1)-Normal achieve 100%; HAR-RV achieves 0%. *Middle panel*: Average absolute VaR 99% (higher = more conservative risk estimate). Student- $t$  models are systematically more conservative than Normal models; HAR-RV is the least conservative of all. *Right panel*: Forecast MAE (lower = more accurate). HAR-RV achieves the lowest MAE; the best in-sample GARCH model (EGARCH(1,1)-Student- $t$ ) ranks third. The figure illustrates the central trade-off of the study: HAR-RV produces the most accurate volatility point forecasts but the worst VaR calibration; Normal GARCH models achieve the best backtesting performance but with less conservative risk estimates than Student- $t$  models.



**Table 18 HAR-RV OLS Regression Results (Training set,  $N = 2,995$  observations after lag construction, January 2010 – December 2021). Dependent variable:  $\widehat{RV}_{t+1} = |r_{t+1}|$ . Heteroskedasticity-robust standard errors.**

\*\*\*  $p < 0.001$ ; n.s. = not significant. Table 7 in the paper body.

| Component                                       | Coefficient | Std. Error | $t$ -stat | $p$ -value | Sig. |
|-------------------------------------------------|-------------|------------|-----------|------------|------|
| $c$ (Intercept)                                 | 0.1377      | 0.0225     | 6.11      | < 0.001    | ***  |
| $\hat{\beta}_d$ (Daily, $RV_{t-1}^{(d)}$ )      | -0.0105     | 0.0221     | -0.48     | 0.635      | n.s. |
| $\hat{\beta}_w$ (Weekly, $RV_{t-1}^{(w)}$ )     | 0.6201      | 0.0415     | 14.96     | < 0.001    | ***  |
| $\hat{\beta}_m$ (Monthly, $RV_{t-1}^{(m)}$ )    | 0.1922      | 0.0415     | 4.63      | < 0.001    | ***  |
| $R^2$                                           | 0.281       |            |           | —          |      |
| $F$ -statistic                                  | 390.2       |            | < 0.001   | ***        |      |
| Persistence ( $\hat{\beta}_w + \hat{\beta}_m$ ) | 0.812       |            |           | —          |      |

The daily component  $\hat{\beta}_d = -0.0105$  is not significant ( $p = 0.635$ ): single-day noise in  $|r_t|$  carries no independent predictive value once weekly and monthly averages are controlled for. The HAR-RV thus effectively reduces to a two-component specification.

Persistence =  $\hat{\beta}_w + \hat{\beta}_m = 0.812$ , comparable to  $\alpha + \beta = 0.990$  in GARCH(1,1)-Student- $t$ .

**Table 19 Diebold-Mariano (1995) Forecast Accuracy Tests: HAR-RV vs. Each GARCH Model (Test set,  $T = 755$ ).  $H_0$ : equal expected forecast loss (two-sided). Negative DM  $\Rightarrow$  HAR-RV superior; Positive DM  $\Rightarrow$  GARCH superior. \*\*\*  $p < 0.001$ ; \*\*  $p < 0.01$ ; \*  $p < 0.05$ .**

| Model (vs. HAR-RV)          | MSE-based ( $\ell^2$ loss) |         |      | MAE-based ( $\ell^1$ loss) |         |      | QLIKE-based (asymmetric) <sup>a</sup> |         |      |
|-----------------------------|----------------------------|---------|------|----------------------------|---------|------|---------------------------------------|---------|------|
|                             | DM                         | $p$     | Sig. | DM                         | $p$     | Sig. | DM                                    | $p$     | Sig. |
| GARCH(1,1)-Normal           | -4.293                     | < 0.001 | ***  | -6.662                     | < 0.001 | ***  | +5.456                                | < 0.001 | ***  |
| GARCH(1,1)-Student- $t$     | -4.617                     | < 0.001 | ***  | -6.685                     | < 0.001 | ***  | +5.373                                | < 0.001 | ***  |
| GJR-GARCH(1,1)-Normal       | -3.129                     | 0.002   | **   | -4.413                     | < 0.001 | ***  | +5.829                                | < 0.001 | ***  |
| GJR-GARCH(1,1)-Student- $t$ | -3.427                     | 0.001   | ***  | -4.204                     | < 0.001 | ***  | +5.611                                | < 0.001 | ***  |
| EGARCH(1,1)-Normal          | -2.510                     | 0.012   | *    | -4.173                     | < 0.001 | ***  | +5.795                                | < 0.001 | ***  |
| EGARCH(1,1)-Student- $t$    | -2.833                     | 0.005   | **   | -3.851                     | < 0.001 | ***  | +5.432                                | < 0.001 | ***  |

<sup>a</sup> QLIKE =  $\ln \hat{\sigma}_t^2 + RV_t^2 / \hat{\sigma}_t^2$  penalises under-prediction more heavily than over-prediction.

MSE & MAE: All DM statistics negative — HAR-RV point-forecast superiority is statistically confirmed at conventional levels.

QLIKE: Ranking **reverses completely** — all GARCH models outperform HAR-RV ( $p < 0.001$ ; DM = +5.37 to +5.83), confirming systematic volatility under-prediction by HAR-RV during high-risk periods and explaining its failure under Basel III backtesting.

HAC variance (Newey-West,  $h = 1$ ); Diebold and Mariano [20].

**Table 20 Expected Shortfall Backtesting: McNeil & Frey (2000) Exceedance Residuals Test at the 99 % Confidence Level (Test set,  $T = 755$ ). On days where the realized return falls below the VaR 99% threshold (violation days), the exceedance residual is  $z_t = r_t - \widehat{\text{ES}}_{99,t}$ .  $H_0: E[z_t] = 0$  (ES correctly specified). Pass criterion:  $p > 0.05$  (one-sample  $t$ -test).**

| Model                       | $N_{\text{viol}}$ | Mean $z$                                | $t$ -stat | $p$ -value | Pass? |
|-----------------------------|-------------------|-----------------------------------------|-----------|------------|-------|
| GARCH(1,1)-Normal           | 13                | -0.1832                                 | -1.1553   | 0.2705     | Yes   |
| GARCH(1,1)-Student- $t$     | 3                 | N/A — insufficient violations ( $< 5$ ) |           |            | —     |
| GJR-GARCH(1,1)-Normal       | 17                | -0.1478                                 | -1.0950   | 0.2897     | Yes   |
| GJR-GARCH(1,1)-Student- $t$ | 3                 | N/A — insufficient violations ( $< 5$ ) |           |            | —     |
| EGARCH(1,1)-Normal          | 13                | -0.3032                                 | -2.0947   | 0.0581     | Yes   |
| EGARCH(1,1)-Student- $t$ ★  | 4                 | N/A — insufficient violations ( $< 5$ ) |           |            | —     |
| HAR-RV                      | 36                | -0.3044                                 | -3.7042   | 0.0007     | No    |

A negative mean  $z$  indicates that actual losses on violation days exceeded the ES forecast, i.e. the model *under-estimated* the expected shortfall on the worst days.

Student- $t$  models generate too few violations at 99 % ( $N < 5$ ) to permit a reliable  $t$ -test; their ES conservatism renders formal ES backtesting uninformative at this level.

HAR-RV fails the ES test ( $p = 0.0007$ ), confirming that its tail risk underestimation extends beyond VaR miscalibration to ES as well. Normal-distribution GARCH models pass, consistent with the Kupiec results in Table 16.

ES backtest method: McNeil and Frey [21] exceedance residuals with one-sample  $t$ -test; minimum  $N_{\text{viol}} = 5$  required.

**Table 21 GARCH Standardised Residual Diagnostics (Training set,  $T = 3,017$ ). Standardised residuals:  $z_t = \hat{\varepsilon}_t / \hat{\sigma}_t$ . ARCH-LM test uses 10 lags;  $H_0$ : no remaining ARCH effect in  $z_t^2$ . Jarque-Bera tests  $H_0$ :  $z_t$  is normally distributed. A well-specified GARCH model should resolve all ARCH effects (ARCH-LM  $p > 0.05$ ) even if  $z_t$  is non-normal (Student- $t$  models are not expected to pass normality).**

| Model                       | Mean $z$ | Std $z$ | Skew   | Ex. Kurt. | ARCH-LM | ARCH $p$ | ARCH res.? | JB $p$    |
|-----------------------------|----------|---------|--------|-----------|---------|----------|------------|-----------|
| GARCH(1,1)-Normal           | -0.054   | 0.998   | -0.688 | 2.136     | 8.901   | 0.5415   | Yes        | $< 0.001$ |
| GARCH(1,1)-Student- $t$     | -0.072   | 0.987   | -0.721 | 2.337     | 8.054   | 0.6235   | Yes        | $< 0.001$ |
| GJR-GARCH(1,1)-Normal       | -0.011   | 1.000   | -0.784 | 2.877     | 7.608   | 0.6670   | Yes        | $< 0.001$ |
| GJR-GARCH(1,1)-Student- $t$ | -0.043   | 1.003   | -0.931 | 4.347     | 6.715   | 0.7521   | Yes        | $< 0.001$ |
| EGARCH(1,1)-Normal          | -0.003   | 1.000   | -0.784 | 2.877     | 5.305   | 0.8699   | Yes        | $< 0.001$ |
| EGARCH(1,1)-Student- $t$ ★  | -0.037   | 1.006   | -0.956 | 4.849     | 7.429   | 0.6844   | Yes        | $< 0.001$ |

**ARCH resolved:** all six GARCH models successfully remove all remaining ARCH effects from the standardised residuals (ARCH-LM  $p > 0.54$  in all cases), confirming adequate variance specification.

**Non-normality:** Jarque-Bera rejects normality for all models ( $p < 0.001$ ), driven by persistent negative skewness and excess kurtosis in  $z_t$ . This is expected and not a specification failure: the Student- $t$  models capture fat tails in-distribution, while the Normal models necessarily leave some non-normality in the standardised residuals, which is why their VaR estimates are less conservative at the 99 % level.

EGARCH(1,1)-Student- $t$  achieves the highest excess kurtosis in  $z_t$  (4.849) and the best ARCH resolution ( $p = 0.684$ ), consistent with its superior AIC/BIC ranking.

**Table 22 Subsample Backtesting Results: Kupiec POF Test Across Market Regimes (Test set, December 2021 – December 2024). The full test period is disaggregated into the 2022 bear market ( $T = 251$ ) and the 2023–24 recovery ( $T = 501$ ). Pass criterion: Kupiec  $p > 0.05$ . Expected violation rates: 5.00% at 95% CL; 1.00% at 99% CL.**

| Model                                                                                  | 95% Confidence Level |         |      |            | 99% Confidence Level |         |      |            |
|----------------------------------------------------------------------------------------|----------------------|---------|------|------------|----------------------|---------|------|------------|
|                                                                                        | Rate                 | Kup $p$ | Pass | $\Delta^a$ | Rate                 | Kup $p$ | Pass | $\Delta^a$ |
| <i>Panel A: Full Test Period (<math>T = 755</math>, December 2021 – December 2024)</i> |                      |         |      |            |                      |         |      |            |
| GARCH(1,1)-Normal                                                                      | 6.49%                | 0.072   | YES  | —          | 1.72%                | 0.071   | YES  | —          |
| GARCH(1,1)-Student- $t$                                                                | 3.05%                | 0.008   | NO   | —          | 0.40%                | 0.058   | YES  | —          |
| GJR-GARCH(1,1)-Normal                                                                  | 7.28%                | 0.007   | NO   | —          | 2.25%                | 0.003   | NO   | —          |
| GJR-GARCH(1,1)-Student- $t$                                                            | 3.18%                | 0.014   | NO   | —          | 0.40%                | 0.058   | YES  | —          |
| EGARCH(1,1)-Normal                                                                     | 6.62%                | 0.051   | YES  | —          | 1.72%                | 0.071   | YES  | —          |
| EGARCH(1,1)-Student- $t$                                                               | 3.05%                | 0.008   | NO   | —          | 0.53%                | 0.154   | YES  | —          |
| HAR-RV                                                                                 | 10.99%               | 0.000   | NO   | —          | 4.77%                | 0.000   | NO   | —          |
| <i>Panel B: Bear Market 2022 (<math>T = 251</math>, January – December 2022)</i>       |                      |         |      |            |                      |         |      |            |
| GARCH(1,1)-Normal                                                                      | 9.16%                | 0.006   | NO   | +2.67      | 2.79%                | 0.020   | NO   | +1.07      |
| GARCH(1,1)-Student- $t$                                                                | 3.98%                | 0.445   | YES  | +0.93      | 0.40%                | 0.276   | YES  | 0.00       |
| GJR-GARCH(1,1)-Normal                                                                  | 10.36%               | 0.001   | NO   | +3.08      | 3.59%                | 0.001   | NO   | +1.34      |
| GJR-GARCH(1,1)-Student- $t$                                                            | 4.78%                | 0.873   | YES  | +1.60      | 0.40%                | 0.276   | YES  | 0.00       |
| EGARCH(1,1)-Normal                                                                     | 10.36%               | 0.001   | NO   | +3.74      | 1.99%                | 0.164   | YES  | +0.27      |
| EGARCH(1,1)-Student- $t$                                                               | 4.78%                | 0.873   | YES  | +1.73      | 0.40%                | 0.276   | YES  | −0.13      |
| HAR-RV                                                                                 | 14.74%               | 0.000   | NO   | +3.75      | 5.98%                | 0.000   | NO   | +1.21      |
| <i>Panel C: Recovery 2023–24 (<math>T = 501</math>, January 2023 – December 2024)</i>  |                      |         |      |            |                      |         |      |            |
| GARCH(1,1)-Normal                                                                      | 5.19%                | 0.847   | YES  | −1.30      | 1.20%                | 0.666   | YES  | −0.52      |
| GARCH(1,1)-Student- $t$                                                                | 2.59%                | 0.007   | NO   | −0.46      | 0.40%                | 0.124   | YES  | 0.00       |
| GJR-GARCH(1,1)-Normal                                                                  | 5.79%                | 0.429   | YES  | −1.49      | 1.60%                | 0.217   | YES  | −0.65      |
| GJR-GARCH(1,1)-Student- $t$                                                            | 2.40%                | 0.003   | NO   | −0.78      | 0.40%                | 0.124   | YES  | 0.00       |
| EGARCH(1,1)-Normal                                                                     | 4.79%                | 0.828   | YES  | −1.83      | 1.60%                | 0.217   | YES  | −0.12      |
| EGARCH(1,1)-Student- $t$                                                               | 2.20%                | 0.001   | NO   | −0.85      | 0.60%                | 0.329   | YES  | +0.07      |
| HAR-RV                                                                                 | 9.18%                | 0.000   | NO   | −1.81      | 4.19%                | 0.000   | NO   | −0.58      |

<sup>a</sup>  $\Delta$  = violation rate in subperiod minus full-test rate (percentage points); positive = more violations than full-test average.

Normal-distribution GARCH models pass in the recovery period but fail during the 2022 bear market, revealing that their full-sample pass rate reflects partial error cancellation across regimes rather than uniform robustness.

Student- $t$  models pass during 2022 (conservative enough) but fail in recovery (over-conservative), consistent with their full-sample 95% failure pattern.

HAR-RV fails in both subperiods at both confidence levels, confirming a structural level bias independent of market regime.

**Table 23 ARIMA Mean-Equation Specification Comparison (Training set,  $T = 3,017$ , January 2010 – December 2021). All specifications estimated by Maximum Likelihood via `statsmodels.tsa.arima`.  $LB_p(10)$ : Ljung-Box portmanteau test  $p$ -value at 10 lags;  $H_0$ : no residual autocorrelation. Lower AIC/BIC = better fit.**

| Specification  | AIC             | BIC      | $LB_p(10)$ | Selected |
|----------------|-----------------|----------|------------|----------|
| Constant mean  | 9,062.85        | 9,074.87 | < 0.001    |          |
| ARIMA(1, 0, 0) | 8,996.81        | 9,014.85 | < 0.001    |          |
| ARIMA(1, 0, 1) | 8,984.62        | 9,008.67 | < 0.001    |          |
| ARIMA(4, 0, 0) | <b>8,972.27</b> | 9,008.34 | < 0.001    | ✓        |

ARIMA(4, 0, 0) selected on lowest AIC (8,972.27), consistent with `auto_arima` stepwise search.

$LB_p < 0.001$  across all specifications reflects weak but economically negligible autocorrelation ( $\max |\hat{\phi}| = 0.137$ ); AR terms purge the innovation sequence prior to GARCH estimation and do not indicate model failure.

Robustness check under a constant mean equation leaves all pass/fail conclusions in Table 17 qualitatively unchanged; mean-equation choice does not drive the main findings.

**Table 24 Restricted GJR-GARCH(1,1)-Student- $t$  Robustness Check: Likelihood Ratio Test of  $H_0: \alpha = 0$  (Training set,  $T = 3,017$ ). The restricted model sets  $\alpha \equiv 0$ , retaining only the leverage term  $\gamma$  and GARCH persistence  $\beta$ .**

|                | Unrestricted | Restricted ( $\alpha \equiv 0$ ) | Difference |
|----------------|--------------|----------------------------------|------------|
| $\hat{\omega}$ | 0.0320***    | 0.0320***                        | 0.0000     |
| $\hat{\alpha}$ | 0.0038       | —                                | —          |
| $\hat{\gamma}$ | 0.3015***    | 0.3028***                        | +0.0013    |
| $\hat{\beta}$  | 0.8214***    | 0.8239***                        | +0.0025    |
| $\hat{\nu}$    | 5.461***     | 5.451***                         | −0.010     |
| Log-likelihood | −3,621.62    | −3,621.66                        | −0.04      |
| AIC            | 7,253.25     | 7,251.33                         | −1.92      |
| BIC            | 7,283.31     | 7,275.37                         | −7.93      |

*Likelihood Ratio Test:  $H_0: \alpha = 0$*

LR statistic 0.078

$p$ -value (df = 1) 0.780

Conclusion Fail to reject  $H_0$  — restriction is not costly

\*\*\*  $p < 0.001$ . Parameters estimated on ARIMA(4, 0, 0) residuals by Maximum Likelihood (`arch` library).

LR statistic = 0.078 ( $p = 0.780$ ): imposing  $\alpha = 0$  does not significantly reduce log-likelihood, and the restricted model achieves lower AIC (−1.92) and BIC (−7.93), confirming that  $\hat{\alpha} = 0.0038$  is not meaningfully different from zero.

The leverage effect is fully captured by  $\hat{\gamma} = 0.303$  ( $p < 0.001$ ), virtually unchanged across specifications ( $\Delta = +0.001$ ), confirming parameter stability. AIC/BIC values differ from Table 12 because this check fits directly on ARIMA residuals rather than jointly estimating mean and variance equations; the substantive conclusion is unaffected.